

板子

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让代码跑起来

编译运行

```
g++ -std=c++23 a.cpp -o a.out  
./a.out
```

对拍器

```
# 请根据实际的C++版本修改-std版本
g++ generator.cpp -o generator.out -std=c++20 -O2 || exit 1
g++ solution.cpp -o solution.out -std=c++20 -O2 || exit 1
g++ brute_force.cpp -o brute_force.out -std=c++20 -O2 || exit 1
mkdir -p testcases
count=1
while true; do
    echo "Running test $count"
    ./generator.out > testcases/input.txt
    timeout 2s ./solution.out < testcases/input.txt > testcases/output_solution.txt
    exit_code=$?
    if [ $exit_code -eq 124 ]; then
        echo -e "\033[31mTime Limit Exceeded on solution\033[0m"
        break
    elif [ $exit_code -eq 139 ]; then
        echo -e "\033[31mRuntime Error (Segmentation fault) on solution\033[0m"
        cat testcases/input.txt
        break
    elif [ $exit_code -eq 134 ]; then
        echo -e "\033[31mRuntime Error (Aborted) on solution\033[0m"
        cat testcases/input.txt
        break
    elif [ $exit_code -eq 136 ]; then
        echo -e "\033[31mRuntime Error (Floating point exception) on solution\033[0m"
        cat testcases/input.txt
        break
    fi
    timeout 2s ./brute_force.out < testcases/input.txt > testcases/output_brute_force.txt
    if [ $exit_code -eq 124 ]; then
        echo -e "\033[31mTime Limit Exceeded on brute force\033[0m"
        break
    fi
    if ! diff -wB testcases/output_solution.txt testcases/output_brute_force.txt > /dev/null; then
        echo -e "\033[31mWrong Answer on test case $count\033[0m"
        echo "Input:"
        cat testcases/input.txt
        echo -e "\nSolution output:"
        cat testcases/output_solution.txt
        echo -e "\nBrute force output:"
        cat testcases/output_brute_force.txt
        break
    fi
    echo -e "\033[32mAccepted\033[0m"
    ((count++))
done
```

Debug宏

```
#ifdef ONLINE_JUDGE
#define debug(...) (void(0))
#else

string trim(string s) {
    int l = 0, r = s.size() - 1;
    while(l <= r && isspace(s[l])) ++l;
    while(r >= l && isspace(s[r])) --r;
    return s.substr(l, r - l + 1);
}

vector<string> split_args(const string &s) {
    vector<string> ret;
    string cur;
    int dep = 0;
    for(char ch : s) {
        if(ch == '(' || ch == '[') {
            ++dep;
            cur += ch;
        } else if(ch == ')' || ch == ']') {
            --dep;
            cur += ch;
        } else if(dep == 0 && ch == ',') {
            ret.push_back(trim(cur));
            cur = "";
        } else {
            cur += ch;
        }
    }
    if(!(trim(cur).empty())) ret.push_back(trim(cur));
    return ret;
}

template<class Tuple, size_t ... I> void print_tuple
    (const vector<string> &names, const Tuple &t, index_sequence<I...>) {
    using expr = int[];
    bool first = true;
    (void)expr{0, ((void)(
        (first ? first = false : (bool)(cerr << ", ")),
        cerr << names[I] << " = " << get<I>(t)
    )), 0)...};
}

template<class ... Args> void debug_helper(const string &s, Args &&... args) {
    auto names = split_args(s);
    auto tp = forward_as_tuple(forward<Args>(args)...);
    print_tuple(names, tp, make_index_sequence<sizeof...(Args)>{});
    cerr << endl;
}

#define debug(...) debug_helper(#__VA_ARGS__, __VA_ARGS__)
#endif
```

数据结构

线段树

```
template<class Info> concept SegInfo = requires(Info a, Info b) {
    Info{};
    { a + b } -> same_as<Info>;
    { a.update(b) } -> same_as<void>;
};

template<SegInfo Info> struct SegTree {
    SegTree() : n(0) {}
    explicit SegTree(int sz) : n(sz), info(sz * 4, Info()) {}
    explicit SegTree(const vector<Info> &v) : n(v.size()), info(v.size() * 4) {
        _build(v, 0, 0, n);
    }
    void assign(int sz) {
        n = sz;
        info.assign(n * 4, Info());
    }
    void assign(const vector<Info> &v) {
        n = v.size();
        info.assign(n * 4, Info());
        _build(v, 0, 0, n);
    }
    Info query(int l, int r) const {
        if(l == r) return Info();
        return _query(l, r, 0, 0, n);
    }
    void update(int x, const Info &v) {
        _update(x, v, 0, 0, n);
    }
private:
    int n;
    vector<Info> info;
    void _build(const vector<Info> &v, int u, int r1, int rr) {
        if(r1 == rr - 1) {
            info[u] = v[r1];
            return;
        }
        int mid = (r1 + rr) / 2, ls = u * 2 + 1, rs = u * 2 + 2;
        _build(v, ls, r1, mid);
        _build(v, rs, mid, rr);
        info[u] = info[ls] + info[rs];
    }
    Info _query(int ql, int qr, int u, int r1, int rr) const {
        if(ql <= r1 && qr >= rr) return info[u];
        int mid = (r1 + rr) >> 1, ls = u * 2 + 1, rs = u * 2 + 2;
        Info res{};
        if(ql < mid) res = res + _query(ql, qr, ls, r1, mid);
        if(qr > mid) res = res + _query(ql, qr, rs, mid, rr);
        return res;
    }
    void _update(int x, const Info &v, int u, int r1, int rr) {
        if(r1 == rr - 1) {
            v.update(info[u]);
        }
    }
};
```

```

        return;
    }
    int mid = (r1 + rr) >> 1, ls = u * 2 + 1, rs = u * 2 + 2;
    if(x < mid) _update(x, v, ls, r1, mid);
    else _update(x, v, rs, mid, rr);
    info[u] = info[ls] + info[rs];
}
};

struct Info {
    Info() {}

}
Info operator+(const Info &i) const {

}
void update(Info &dst) const {

}
};

template<class Info, class Tag> concept SegInfoTag = requires(Info a, Info b, Tag c, Tag d, int l, int r) {
    Info{};
    Tag{};
    { a + b } -> same_as<Info>;
    { c.apply(a, l, r) } -> same_as<void>;
    { c.apply(d, l, r) } -> same_as<void>;
    { c.empty() } -> same_as<bool>;
    { c.clear() } -> same_as<void>;
} && !is_same_v<Info, bool>;

template<class Info, class Tag> requires(SegInfoTag<Info, Tag>) struct LazySegTree {
    LazySegTree() : n(0) {}
    explicit LazySegTree(int n_) : n(n_), info(4 * n_), tag(4 * n_) {}
    explicit LazySegTree(const vector<Info> &v) : n(v.size()), info(4 * v.size()), tag(4 * v.size()) {
        _build(v, 0, 0, n);
    }
    void assign(int n_) {
        n = n_;
        info.assign(4 * n_, Info{});
        tag.assign(4 * n_, Tag{});
    }
    void assign(const vector<Info> &v) {
        n = v.size();
        info.assign(4 * v.size(), Info{});
        tag.assign(4 * v.size(), Tag{});
        _build(v, 0, 0, n);
    }
    Info query(int l, int r) {
        return _query(l, r, 0, 0, n);
    }
    void update(int l, int r, const Tag &t) {
        _update(l, r, t, 0, 0, n);
    }
private:
    int n;
    vector<Info> info;
    vector<Tag> tag;

```

```

void _build(const vector<Info> &v, int u, int rl, int rr) {
    if(rr - rl == 1) {
        info[u] = v[rl];
        return;
    }
    int mid = (rl + rr) / 2, ls = u * 2 + 1, rs = u * 2 + 2;
    _build(v, ls, rl, mid); _build(v, rs, mid, rr);
    info[u] = info[ls] + info[rs];
}

void _pushdown(int u, int rl, int rr) {
    if(tag[u].empty()) return;
    int mid = (rl + rr) / 2, ls = u * 2 + 1, rs = u * 2 + 2;
    tag[u].apply(info[ls], rl, mid);
    tag[u].apply(info[rs], mid, rr);
    tag[u].apply(tag[ls], rl, mid);
    tag[u].apply(tag[rs], mid, rr);
    tag[u].clear();
}

Info _query(int ql, int qr, int u, int rl, int rr) {
    if(ql <= rl && qr >= rr) return info[u];
    _pushdown(u, rl, rr);
    Info ret{};
    int mid = (rl + rr) / 2, ls = u * 2 + 1, rs = u * 2 + 2;
    if(ql < mid) ret = ret + _query(ql, qr, ls, rl, mid);
    if(qr > mid) ret = ret + _query(ql, qr, rs, mid, rr);
    return ret;
}

void _update(int ul, int ur, const Tag &t, int u, int rl, int rr) {
    if(ul <= rl && ur >= rr) {
        t.apply(info[u], rl, rr);
        t.apply(tag[u], rl, rr);
        return;
    }
    _pushdown(u, rl, rr);
    int mid = (rl + rr) / 2, ls = u * 2 + 1, rs = u * 2 + 2;
    if(ul < mid) _update(ul, ur, t, ls, rl, mid);
    if(ur > mid) _update(ul, ur, t, rs, mid, rr);
    info[u] = info[ls] + info[rs];
}

};

struct Info {
    Info() {}

}

Info operator+(const Info &i) const {

}

};

struct Tag {
    Tag() {}

}

bool empty() const {

}

void clear() {

```



```
    }  
    void apply(Info &dst, int l, int r) const {  
  
    }  
    void apply(Tag &dst, int l, int r) const {  
  
    }  
};
```

权值线段树

```
// value segment tree
struct VST {
    explicit VST(int maxn) : sum((maxn + 1) * 4), n(maxn + 1) {}
    VST() : n(0) {}
    void assign(int maxn) {
        sum.assign((maxn + 1) * 4, 0);
        n = maxn + 1;
    }
    void insert(int x) {
        _update(x, 1, 0, 0, n);
    }
    void erase(int x) {
        _update(x, -1, 0, 0, n);
    }
    int qpre(int x) const {
        int rk = qrv(x);
        return qvr(rk - 1);
    }
    int qsuc(int x) const {
        int rk = qrv(x + 1);
        return qvr(rk);
    }
    int qrv(int x) const {
        return _query(0, x, 0, 0, n);
    }
    int qvr(int k) const {
        return _qvr(k, 0, 0, n);
    }
    int size() const {
        return sum[0];
    }
    int qcnt(int x) const {
        return _query(x, x + 1, 0, 0, n);
    }
    int qrange_cnt(int l, int r) const {
        return _query(l, r, 0, 0, n);
    }
private:
    vector<int> sum;
    int n;
    void _update(int x, int dv, int rt, int r1, int rr) {
        if(rr - r1 == 1) {
            sum[rt] += dv;
            return;
        }
        int mid = (r1 + rr) >> 1, lson = rt * 2 + 1, rson = rt * 2 + 2;
        if(x < mid) {
            _update(x, dv, lson, r1, mid);
        } else {
            _update(x, dv, rson, mid, rr);
        }
        sum[rt] = sum[lson] + sum[rson];
    }
    ll _query(int ql, int qr, int rt, int r1, int rr) const {
        if(ql <= r1 && qr >= rr)
```

```

        return sum[rt];
    int mid = (r1 + rr) >> 1, lson = rt * 2 + 1, rson = rt * 2 + 2;
    ll ans = 0;
    if(q1 < mid) {
        ans += _query(q1, qr, lson, r1, mid);
    }
    if(qr > mid) {
        ans += _query(q1, qr, rson, mid, rr);
    }
    return ans;
}

int _qvr(int k, int rt, int r1, int rr) const {
    if(rr - r1 == 1)
        return r1;
    int mid = (r1 + rr) >> 1, lson = rt * 2 + 1, rson = rt * 2 + 2;
    if(k < sum[lson]) {
        return _qvr(k, lson, r1, mid);
    } else {
        return _qvr(k - sum[lson], rson, mid, rr);
    }
}

};

```

主席树

```
// persistent segment tree
struct PST {
    const int n;
    explicit PST(int n_) : n(n_) {
        nodes.reserve(2 * n * (int)log2(n));
        vers.reserve(n + 1);
        vers.push_back(_build(0, n - 1));
    }
    void update(int k) {
        vers.push_back(_update(k, vers.back(), 0, n - 1));
    }
    // 这里为解决静态区间第k小问题设计, 传入k, l, r
    int query(int k, int ver1, int ver2) const {
        return _query(k, vers[ver1 - 1], vers[ver2], 0, n - 1);
    }
private:
    struct Node {
        int l, r;
        int cnt;
    };
    vector<Node> nodes;
    vector<int> vers;
    int _build(int l, int r) {
        int ret = nodes.size();
        if(l == r) {
            nodes.emplace_back(-1, -1, 0);
            return ret;
        }
        nodes.emplace_back(-1, -1, 0);
        int mid = (l + r) >> 1;
        nodes[ret].l = _build(l, mid);
        nodes[ret].r = _build(mid + 1, r);
        return ret;
    }
    int _update(int k, int root, int l, int r) {
        int ret = nodes.size();
        nodes.emplace_back(nodes[root]);
        nodes[ret].cnt++;
        if(l == r) {
            return ret;
        }
        int mid = (l + r) >> 1;
        if(k <= mid) {
            nodes[ret].l = _update(k, nodes[root].l, l, mid);
        } else {
            nodes[ret].r = _update(k, nodes[root].r, mid + 1, r);
        }
        return ret;
    }
    int _query(int k, int root1, int root2, int l, int r) const {
        if(l == r) return l;
        int mid = (l + r) >> 1;
        int cnt1 = nodes[nodes[root2].l].cnt - nodes[nodes[root1].l].cnt;
        if(k <= cnt1) {
            return _query(k, nodes[root1].l, nodes[root2].l, l, mid);
        }
    }
};
```

```

        } else {
            return _query(k - cnt1, nodes[root1].r, nodes[root2].r, mid + 1, r);
        }
    }
};

```

树状数组（单点、区间）

```

// 单点
struct Fenwick {
    explicit Fenwick(int n) : n(n), nums(n + 1, 0) {}
    ll query(int x) const {
        ll ans = 0;
        for(; x; x -= lbit(x)) {
            ans += nums[x];
        }
        return ans;
    }
    void update(int x, ll v) {
        for(; x <= n; x += lbit(x)) {
            nums[x] += v;
        }
    }
private:
    vector<ll> nums;
    int n;
    static int lbit(int x) {
        return x & -x;
    }
};

// 区间
struct RangeFenwick {
    const int n;
    RangeFenwick(int n)
        : n(n), f1(n), f2(n) {}
    void update(int l, int r, ll v) {
        _update(l, v);
        _update(r + 1, -v);
    }
    ll query(int l, int r) const {
        return _query(r) - _query(l - 1);
    }
private:
    Fenwick f1, f2;
    void _update(int x, ll v) {
        f1.update(x, v);
        f2.update(x, v * (x - 1));
    }
    ll _query(int x) const {
        return f1.query(x) * x - f2.query(x);
    }
};

```

差分数组

使用差分数组区间加等差数列

```
// 二次差分
vector<ll> diff(n + 3, 0);
// [1,1]第i项加i
diff[1] += val;
diff[1 + 1] -= val;
// [r+1,l+r]第i项减i-r
diff[r + 1] -= val;
diff[1 + r + 1] += val;
// 还原
for(int _ = 0; _ < 2; ++_) {
    for(int i = 1; i < n + 3; ++i) {
        diff[i] += diff[i - 1];
    }
}
```

树状数组在线处理

```
struct ArithmeticFenwick {
    explicit ArithmeticFenwick(int n)
        : f1(n + 2), f2(n + 2) {}
    // a*idx+b
    void update(int l, int r, ll a, ll b) {
        f1.update(l, b);
        f1.update(r + 1, -b - a * (r - l + 1));
        f2.update(l, a);
        f2.update(r + 1, -a);
    }
    ll query(ll idx) const {
        return f1.query(idx) + idx * f2.query(idx);
    }
private:
    Fenwick f1, f2;
};
```

分块

```
struct Block {
    Block() : l(-1), r(-1) {}
    // 注意左闭右开
    Block(int l, int r, const vector<ll> &nums_)
        : nums(r - l), lazy(0), sum(0), l(l), r(r) {
        for(int i = l; i < r; ++i) {
            nums[i - l] = nums_[i];
            sum += nums[i - l];
        }
    }
    // 注意左闭右开
    void update(int ul, int ur, ll dval) {
        if(ul > r || ur < l) {
            return;
        }
        if(ul <= l && ur >= r) {
            lazy += dval;
            sum += dval * (r - l);
            return;
        }
        for(int i = max(ul, l); i < min(ur, r); ++i) {
            nums[i - l] += dval;
        }
        sum += dval * (min(ur, r) - max(ul, l));
    }
    // 注意左闭右开
    ll query(int ql, int qr) const {
        if(ql >= r || qr <= l) {
            return 0;
        }
        if(ql <= l && qr >= r) {
            return sum;
        }
        ll ans = lazy * (min(qr, r) - max(ql, l));
        for(int i = max(ql, l); i < min(qr, r); ++i) {
            ans += nums[i - l];
        }
        return ans;
    }
private:
    vector<ll> nums;
    ll lazy;
    ll sum;
    int l, r;
};

struct BlockedArray {
    explicit BlockedArray(const vector<ll> &nums) {
        int r = nums.size();
        int len = sqrt(r + 0.5);
        int size = (r + len - 1) / len;
        blocks.reserve(size);
        for(int i = 0; i < size; ++i) {
            int ll = i * len, rr = min((i + 1) * len, r);
            blocks.emplace_back(ll, rr, nums);
        }
    }
};
```

```
    }  
    void update(int l, int r, ll d) {  
        for(auto &blk : blocks) {  
            blk.update(l, r, d);  
        }  
    }  
    ll query(int l, int r) const {  
        ll ans = 0;  
        for(const auto &blk : blocks) {  
            ans += blk.query(l, r);  
        }  
        return ans;  
    }  
    vector<Block> blocks;  
};
```


并查集

// 按秩合并

```
struct DSU {
    explicit DSU(int n) : fa(n), rk(n, 1) {
        iota(fa.begin(), fa.end(), 0);
    }
    int find(int x) {
        if(fa[x] != x) {
            fa[x] = find(fa[x]);
        }
        return fa[x];
    }
    bool is_connected(int x, int y) {
        return find(x) == find(y);
    }
    void connect(int x, int y) {
        x = find(x);
        y = find(y);
        if(x == y) return;
        if(rk[x] < rk[y]) {
            swap(x, y);
        }
        if(rk[x] == rk[y]) {
            rk[x]++;
        }
        fa[y] = x;
    }
private:
    vector<int> fa, rk;
};
```

// 随机合并

```
struct DSU_Random {
    explicit DSU_Random(int n) : fa(n) {
        iota(fa.begin(), fa.end(), 0);
    }
    int find(int x) {
        if(fa[x] != x) {
            fa[x] = find(fa[x]);
        }
        return fa[x];
    }
    bool is_connected(int x, int y) {
        return find(x) == find(y);
    }
    void connect(int x, int y) {
        int fx = find(x), fy = find(y);
        if(rnd() % 2) {
            fa[fx] = fy;
        } else {
            fa[fy] = fx;
        }
    }
private:
    static inline mt19937 rnd{ (unsigned int)chrono::steady_clock::now()
        .time_since_epoch().count() };
};
```

```
vector<int> fa;
```

```
};
```

无旋Treap

```
constexpr int inf = 0x3f3f3f3f;
struct Treap {
    Treap() : rt(nullptr) {}
    ~Treap() {
        if(rt) {
            _dtor(rt);
        }
    }
    void insert(int v) {
        auto tmp = _sval(rt, v);
        auto l = _sval(tmp.first, v - 1);
        Node *node = l.second;
        if(!l.second) {
            node = new Node(v);
        } else {
            l.second->cnt++;
            l.second->usize();
        }
        Node *lc = _merge(l.first, node);
        rt = _merge(lc, tmp.second);
    }
    void erase(int v) {
        auto tmp = _sval(rt, v);
        auto l = _sval(tmp.first, v - 1);
        if(l.second->cnt > 1) {
            l.second->cnt--;
            l.second->usize();
            l.first = _merge(l.first, l.second);
        } else {
            if(tmp.first == l.second) {
                tmp.first = nullptr;
            }
            delete l.second;
            l.second = nullptr;
        }
        rt = _merge(l.first, tmp.second);
    }
    // 注意0-based
    int qrv(int v) const {
        return _qrv(rt, v);
    }
    // 注意0-based
    int qvr(int r) const {
        return _qvr(rt, r);
    }
    // 注意是小于的, 不是大于等于的
    int lower_bound(int v) const {
        auto tmp = _sval(rt, v - 1);
        int ret = -inf;
        if(tmp.first) {
            ret = _qvr(tmp.first, tmp.first->size - 1);
        }
        rt = _merge(tmp.first, tmp.second);
        return ret;
    }
}
```

```

int upper_bound(int v) const {
    auto tmp = _sval(rt, v);
    int ret = -inf;
    if(tmp.second) {
        ret = _qvr(tmp.second, 0);
    }
    rt = _merge(tmp.first, tmp.second);
    return ret;
}

private:
struct Node {
    int val;
    int cnt;
    int size;
    int prio;
    Node *left, *right;
    static inline random_device rnd{};
    Node(int v)
        : val(v), cnt(1), size(1), prio(rnd()), left(nullptr), right(nullptr) {}
    void usize() {
        size = cnt;
        if(left) {
            size += left->size;
        }
        if(right) {
            size += right->size;
        }
    }
};

mutable Node *rt;
static void _dtor(Node *ptr) {
    if(ptr->left) {
        _dtor(ptr->left);
        ptr->left = nullptr;
    }
    if(ptr->right) {
        _dtor(ptr->right);
        ptr->right = nullptr;
    }
    delete ptr;
}

static pair<Node *, Node *> _sval(Node *const ptr, int key) {
    if(!ptr) {
        return { nullptr, nullptr };
    }
    if(ptr->val <= key) {
        auto tmp = _sval(ptr->right, key);
        ptr->right = tmp.first;
        ptr->usize();
        return { ptr, tmp.second };
    } else {
        auto tmp = _sval(ptr->left, key);
        ptr->left = tmp.second;
        ptr->usize();
        return { tmp.first, ptr };
    }
}

```

```

static tuple<Node *, Node *, Node *> _srnk(Node *const ptr, int rnk) {
    if(!ptr) return { nullptr, nullptr, nullptr };
    int lsize = (ptr->left ? ptr->left->size : 0);
    if(rnk < lsize) {
        auto [lptr, mptr, rptr] = _srnk(ptr->left, rnk);
        ptr->left = rptr;
        ptr->usize();
        return { lptr, mptr, ptr };
    } else if(rnk < lsize + ptr->cnt) {
        Node *lptr = ptr->left;
        Node *rptr = ptr->right;
        ptr->left = ptr->right = nullptr;
        return { lptr, ptr, rptr };
    } else {
        auto [lptr, mptr, rptr] = _srnk(ptr->right, rnk - lsize - ptr->cnt);
        ptr->right = lptr;
        ptr->usize();
        return { ptr, mptr, rptr };
    }
}

static Node *_merge(Node *const u, Node *const v) {
    if(!u) return v;
    if(!v) return u;
    if(u->prio < v->prio) {
        u->right = _merge(u->right, v);
        u->usize();
        return u;
    } else {
        v->left = _merge(u, v->left);
        v->usize();
        return v;
    }
}

int _qrv(Node *const ptr, int val) const {
    auto [l, r] = _sval(ptr, val - 1);
    int ret = (l ? l->size : 0);
    rt = _merge(l, r);
    return ret;
}

int _qvr(Node *const ptr, int r) const {
    auto [lptr, mptr, rptr] = _srnk(ptr, r);
    int ret = mptr->val;
    rt = _merge(_merge(lptr, mptr), rptr);
    return ret;
}
};

```

树的倍增算法

```
struct BinaryLift {
    explicit BinaryLift(int n) : tree(n), anc(n, vector<int>(LOG, -1)), depth(n) {}
    void addedge(int u, int v) {
        tree[u].push_back(v);
        tree[v].push_back(u);
    }
    void build(int root) {
        dfs(root, -1);
    }
    int lca(int u, int v) const {
        if(depth[u] < depth[v]) swap(u, v);
        for(int k = LOG - 1; k >= 0; --k) {
            if(anc[u][k] != -1) {
                if(depth[anc[u][k]] >= depth[v]) {
                    u = anc[u][k];
                }
            }
        }
        if(u == v) return u;
        for(int k = LOG - 1; k >= 0; --k) {
            if(anc[u][k] != anc[v][k]) {
                u = anc[u][k];
                v = anc[v][k];
            }
        }
        return anc[u][0];
    }
    int kth_ancestor(int x, int k) const {
        for(int i = 0; i < LOG; ++i) {
            if((k >> i) & 1) {
                x = anc[x][i];
                if(x == -1) return -1;
            }
        }
        return x;
    }
    vector<vector<int>> tree;
private:
    static constexpr int LOG = 20;
    vector<vector<int>> anc;
    vector<int> depth;
    void dfs(int root, int fa) {
        depth[root] = fa != -1 ? depth[fa] + 1 : 0;
        anc[root][0] = fa;
        for(int k = 1; k < LOG; ++k) {
            if(anc[root][k - 1] == -1) {
                anc[root][k] = -1;
            } else {
                anc[root][k] = anc[anc[root][k - 1]][k - 1];
            }
        }
        for(int v : tree[root]) {
            if(v != fa) {
                dfs(v, root);
            }
        }
    }
}
```

```
}  
}  
};
```

树链剖分

```
struct HLDInfo {
    ll val;
    explicit HLDInfo(ll v) : val(v) {}
    HLDInfo() : val(0) {}
    void update(HLDInfo &dst) const {
        dst.val = val;
    }
    HLDInfo operator+(const HLDInfo &x) const {
        return HLDInfo(val + x.val);
    }
};

struct Node {
    ll w;
    vector<int> e;
};

struct HLD {
    explicit HLD(const vector<Node> &g, int r = 0)
        : nodes(g.size()), nfd(g.size()) {
        dfs1(g, r, -1);
        int now_index = 0;
        dfs2(g, r, -1, now_index, r);
        vector<HLDInfo> nums(g.size());
        for(int i = 0; i < g.size(); ++i) {
            nums[nodes[i].dfn] = HLDInfo{g[i].w};
        }
        seg.assign(nums);
    }

    int lca(int u, int v) const {
        while(nodes[u].toc != nodes[v].toc) {
            if(nodes[nodes[u].toc].dep < nodes[nodes[v].toc].dep) {
                v = nodes[nodes[v].toc].fa;
            } else {
                u = nodes[nodes[u].toc].fa;
            }
        }
        return nodes[u].dep < nodes[v].dep ? u : v;
    }

    void modify(int x, ll v) {
        seg.update(nodes[x].dfn, HLDInfo{v});
    }

    ll query_sum(int u, int v) const {
        int lca_node = lca(u, v);
        auto query_to_lca = [&](int point) -> ll {
            ll ans = 0;
            for(; nodes[point].toc != nodes[lca_node].toc;
                point = nodes[nodes[point].toc].fa) {
                int pos1 = nodes[point].dfn;
                int pos2 = nodes[nodes[point].toc].dfn;
                ans += seg.query(pos2, pos1 + 1).val;
            }
            int pos1 = nodes[lca_node].dfn;
            int pos2 = nodes[point].dfn;
            ans += seg.query(pos1 + 1, pos2 + 1).val;
            return ans;
        };
    }
};
```



```

    };
    ll ans = query_to_lca(u);
    ans += query_to_lca(v);
    int lca_pos = nodes[lca_node].dfn;
    ans += seg.query(lca_pos, lca_pos + 1).val;
    return ans;
}

private:
    struct _Node {
        int dep, fa, toc, dfn, sz, hs;
    };

    void dfs1(const vector<Node> &g, int x, int fa) {
        if(fa == -1) {
            nodes[x].dep = 0;
        } else {
            nodes[x].dep = nodes[fa].dep + 1;
        }
        nodes[x].fa = fa;
        nodes[x].sz = 1;
        nodes[x].hs = -1;
        for(int i = 0; i < g[x].e.size(); ++i) {
            int next = g[x].e[i];
            if(next == fa) continue;
            dfs1(g, next, x);
            nodes[x].sz += nodes[next].sz;
            if(nodes[x].hs == -1 ||
                nodes[nodes[x].hs].sz < nodes[next].sz) {
                nodes[x].hs = next;
            }
        }
    }

    void dfs2(const vector<Node> &g, int x, int fa, int &dfn, int toc) {
        nodes[x].dfn = dfn++;
        nodes[x].toc = toc;
        if(nodes[x].hs != -1) {
            dfs2(g, nodes[x].hs, x, dfn, toc);
            for(int i = 0; i < g[x].e.size(); ++i) {
                int next = g[x].e[i];
                if(next != nodes[x].hs && next != fa) {
                    dfs2(g, next, x, dfn, next);
                }
            }
        }
    }

    vector<_Node> nodes;
    vector<int> nfd;
    SegTree<HLDInfo> seg;
};

```

动态BitSet

```
struct DynamicBitSet {
    explicit DynamicBitSet(int n = 0) : nums((n + 63) >> 6, 0), sz(n) {}
    void resize(int n) {
        nums.resize(n);
        sz = n;
    }
    bool getbit(int x) const {
        int u = x >> 6;
        int v = x - (u << 6);
        return ((nums[u] >> v) & 1) == 1;
    }
    void setbit(int x, bool val) {
        int u = x >> 6;
        int v = x - (u << 6);
        if(!val) {
            ull d = (ull)-1;
            d ^= (1ull << v);
            nums[u] &= d;
        } else {
            nums[u] |= (1ull << v);
        }
    }
    DynamicBitSet &operator&=(const DynamicBitSet &other) {
        for(int i = 0; i < nums.size(); ++i) {
            nums[i] &= other.nums[i];
        }
        return *this;
    }
    DynamicBitSet operator&(const DynamicBitSet &other) const {
        DynamicBitSet ret = *this;
        ret &= other;
        return ret;
    }
    DynamicBitSet &operator|=(const DynamicBitSet &other) {
        for(int i = 0; i < nums.size(); ++i) {
            nums[i] |= other.nums[i];
        }
        return *this;
    }
    DynamicBitSet operator|(const DynamicBitSet &other) const {
        DynamicBitSet ret = *this;
        ret |= other;
        return ret;
    }
    DynamicBitSet &operator^=(const DynamicBitSet &other) {
        for(int i = 0; i < nums.size(); ++i) {
            nums[i] ^= other.nums[i];
        }
        return *this;
    }
    DynamicBitSet operator^(const DynamicBitSet &other) const {
        DynamicBitSet ret = *this;
        ret ^= other;
        return ret;
    }
}
```

```

    bool allzero() const {
        for(ull i : nums) {
            if(i != 0ull) return false;
        }
        return true;
    }
    int size() const {return sz;}
private:
    vector<ull> nums;
    int sz;
};

```

ST表

```

template<class T> struct Sparse {
    using func_type = function<T(T, T)>;
    Sparse(const vector<T> &vec, func_type fn) : func(fn) {
        int n = vec.size();
        _log.resize(n + 1);
        _log[0] = -1;
        for(int i = 1; i <= n; ++i) {
            _log[i] = _log[i >> 1] + 1;
        }
        int k = _log[n] + 1;
        table.resize(n, vector<T>(k));
        for(int i = 0; i < n; ++i) {
            table[i][0] = vec[i];
        }
        for(int j = 1; j < k; ++j) {
            int step = 1 << (j - 1);
            for(int i = 0; i + step < n; ++i) {
                table[i][j] = func(table[i][j - 1], table[i + step][j - 1]);
            }
        }
    }
    T query(int l, int r) const {
        int len = r - l + 1;
        int j = _log[len];
        return func(table[l][j], table[r - (1 << j) + 1][j]);
    }
private:
    vector<vector<T>> table;
    vector<int> _log;
    func_type func;
};

```

01-Trie

```
struct Trie01 {
    Trie01() : nodes(1) {}
    void insert(ll val) {
        int now = 0;
        for(ll i = 62; i >= 0; --i) {
            ll flag = (val >> i) & 1;
            if(nodes[now].nxt[flag] == -1) {
                nodes[now].nxt[flag] = nodes.size();
                nodes.emplace_back();
            }
            now = nodes[now].nxt[flag];
        }
    }
    ll qmax_xor(ll val) const {
        int now = 0;
        ll ans = 0;
        for(ll i = 62; i >= 0; --i) {
            ll flag = (val >> i) & 1;
            if(nodes[now].nxt[1 ^ flag] != -1) {
                ans += (1ll << i);
                now = nodes[now].nxt[1 ^ flag];
            } else {
                now = nodes[now].nxt[flag];
            }
        }
        return ans;
    }
private:
    static constexpr int height = 63;
    struct Node {
        int nxt[2];
        Node() {
            nxt[0] = nxt[1] = -1;
        }
    };
    vector<Node> nodes;
};
```

CDQ分治

```
struct Data {
    int x, y, z;
    int cnt;
    int ans;
    Data(int x_, int y_, int z_) : x(x_), y(y_), z(z_), cnt(1), ans(-1) {}
};

vector<int> threed_partial(int n, int k, vector<Data> &_datas) {
    Fenwick fwk(k);
    sort(_datas.begin(), _datas.end() - 1, [](const Data &l, const Data &r) {
        if(l.x != r.x) return l.x < r.x;
        if(l.y != r.y) return l.y < r.y;
        return l.z < r.z;
    });
    // 如果值域很大, 考虑离散化数据
    vector<Data> datas;
    datas.reserve(n);
    int cnt = 0;
    for(int i = 0; i < n; ++i) {
        ++cnt;
        if((_datas[i].x != _datas[i + 1].x) ||
            (_datas[i].y != _datas[i + 1].y) ||
            (_datas[i].z != _datas[i + 1].z)) {
            datas.emplace_back(_datas[i].x, _datas[i].y, _datas[i].z);
            cnt = 0;
        }
    }
    int m = datas.size();
    auto cdq = [&](auto &&cdq, int l, int r) -> void {
        if(r - l < 2) {
            return;
        }
        int mid = (l + r) >> 1;
        cdq(cdq, l, mid);
        cdq(cdq, mid, r);
        sort(datas.begin() + l, datas.begin() + mid, [](const Data &l, const Data &r) {
            if(l.y != r.y) return l.y < r.y;
            return l.z < r.z;
        });
        sort(datas.begin() + mid, datas.begin() + r, [](const Data &l, const Data &r) {
            if(l.y != r.y) return l.y < r.y;
            return l.z < r.z;
        });
        int j = l;
        for(int i = mid; i < r; ++i) {
            while(datas[j].y <= datas[i].y && j < mid) {
                fwk.update(datas[j].z, datas[j].cnt);
                ++j;
            }
            datas[i].ans += fwk.query(datas[i].z);
        }
        for(int k = l; k < j; ++k) {
            fwk.update(datas[k].z, -datas[k].cnt);
        }
    };
    cdq(cdq, 0, m);
}
```

```
vector<int> ans(n, 0);  
for(int i = 0; i < m; ++i) {  
    ans[datas[i].ans + datas[i].cnt - 1] += datas[i].cnt;  
}  
return ans;  
}
```

莫队

```
struct Query {
    int idx;
    // 左闭右开
    int l, r;
    int ans;
};

// 查询区间内满足元素出现次数等于自身的数的个数
void mo_algo(vector<Query> &queries, const vector<int> &nums) {
    int n = nums.size();
    int len = sqrt(n);
    sort(queries.begin(), queries.end(), [&](const Query &q1, const Query &q2) {
        int atl = q1.l / len, atr = q1.r / len;
        if(atl != atr) {
            return atl < atr;
        }
        return (bool)((atl % 2 == 0) ^ (q1.r < q2.r));
    });
    int l = 0, r = 0, ans = 0;
    vector<int> count(n + 5, 0);
    auto add = [&](int x) {
        if(x >= n + 5) return;
        if(count[x] == x) --ans;
        count[x]++;
        if(count[x] == x) ++ans;
    };
    auto remove = [&](int x) {
        if(x >= n + 5) return;
        if(count[x] == x) --ans;
        count[x]--;
        if(count[x] == x) ++ans;
    };
    for(auto &q : queries) {
        // 抄板子时注意顺序，先加再更新还是先更新再加
        while(l > q.l) {
            --l;
            add(nums[l]);
        }
        while(l < q.l) {
            remove(nums[l]);
            ++l;
        }
        while(r < q.r) {
            add(nums[r]);
            ++r;
        }
        while(r > q.r) {
            --r;
            remove(nums[r]);
        }
        q.ans = ans;
    }
    sort(queries.begin(), queries.end(), [](const Query &l, const Query &r) {
        return l.idx < r.idx;
    });
}
```

```
});
```

```
}
```


单调栈、单调队列

// 单调栈: [1, r), 且有相同元素时, 左边取最左端, 右边取下一个自己

```
vector<pair<int, int>> max_range(const vector<int> &nums) {
    int n = nums.size();
    vector ret(n, pair(-1, -1));
    vector<pair<int, int>> stk; // 单调栈
    ret[0].first = 0;
    stk.emplace_back(nums[0], 0);
    for(int i = 1; i < n; ++i) {
        while(!stk.empty() && stk.back().first <= nums[i]) {
            ret[stk.back().second].second = i;
            stk.pop_back();
        }
        ret[i].first = stk.empty() ? 0 : stk.back().second + 1;
        stk.emplace_back(nums[i], i);
    }
    for(int i = n - 1; i >= 0; --i) {
        if(ret[i].second == -1) {
            ret[i].second = n;
        }
    }
    return ret;
}
```

// 可以有相同元素

```
vector ret(n, pair(0, n));
vector<int> stk;
for(int i = 0; i < n; ++i) {
    while(!stk.empty() && nums[stk.back()] <= nums[i]) {
        stk.pop_back();
    }
    if(!stk.empty()) {
        ret[i].first = stk.back() + 1;
    }
    stk.push_back(i);
}
stk.clear();
for(int i = n - 1; i >= 0; --i) {
    while(!stk.empty() && nums[stk.back()] <= nums[i]) {
        stk.pop_back();
    }
    if(!stk.empty()) {
        ret[i].second = stk.back();
    }
    stk.push_back(i);
}
```

// 最大滑动窗口: 单调队列

```
vector<int> max_sliding_window(const vector<int> &nums, int k) {
    int n = nums.size();
    vector<int> ret(n - k + 1);
    vector<pair<int, int>> que;
    for(int i = 0; i < k - 1; ++i) {
        while(!que.empty() && que.back().first < nums[i]) {
            que.pop_back();
        }
        que.emplace_back(nums[i], i);
    }
    for(int i = k - 1; i < n; ++i) {
        while(!que.empty() && que.back().first <= nums[i]) {
            que.pop_back();
        }
        que.emplace_back(nums[i], i);
    }
    for(int i = 0; i < n - k + 1; ++i) {
        ret[i] = que.front().first;
        que.pop_front();
    }
    return ret;
}
```

```

int st = 0;
for(int i = 0; i <= n - k; ++i) {
    int j = i + k - 1;
    if(que.size() > st && que[st].second < i) {
        ++st;
    }
    while(!que.empty() && que.back().first < nums[j]) {
        que.pop_back();
    }
    st = min(st, (int)que.size());
    que.emplace_back(nums[j], j);
    ret[i] = que[st].first;
}
return ret;
}

```

图论

Dijkstra

```

constexpr ll inf = 0x3f3f3f3f3f3f3f3f;
vector<ll> dijkstra(vector<vector<pair<int, ll>>> &graph, int start) {
    int v = graph.size();
    vector<ll> dist(v, inf);
    dist[start] = 0;
    vector<bool> visited(v, false);
    priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>> pq;
    for(auto [vtx, w] : graph[start]) {
        dist[vtx] = w;
        pq.emplace(w, vtx);
    }
    while(!pq.empty()) {
        auto [w, vtx] = pq.top();
        pq.pop();
        if(visited[vtx]) continue;
        visited[vtx] = true;
        for(auto [vt, ww] : graph[vtx]) {
            if(!visited[vt]) {
                if(dist[vt] > dist[vtx] + ww) {
                    dist[vt] = dist[vtx] + ww;
                    pq.emplace(dist[vt], vt);
                }
            }
        }
    }
    return dist;
}

```

Floyd

```
// 操作后, graph就变成了最短路
void floyd(vector<vector<ll>> &graph) {
    for(int k = 0; k < graph.size(); ++k) {
        for(int i = 0; i < graph.size(); ++i) {
            for(int j = 0; j < graph.size(); ++j) {
                if(graph[i][j] > graph[i][k] + graph[k][j]) {
                    graph[i][j] = graph[i][k] + graph[k][j];
                }
            }
        }
    }
}
```

SPFA

```
constexpr ll inf = 0x3f3f3f3f3f3f3f3f;
// 返回空vector说明有负环
vector<ll> spfa(const vector<vector<pair<int, ll>>> &graph, int start) {
    int n = graph.size();
    vector<ll> dist(n, inf);
    vector<int> cnt(n, 0);
    vector<bool> inq(n, false);
    queue<int> q;
    q.push(start);
    dist[start] = 0;
    inq[start] = true;
    while(!q.empty()) {
        int u = q.front();
        q.pop();
        inq[u] = false;
        for(auto [v, w] : graph[u]) {
            if(dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                if(!inq[v]) {
                    inq[v] = true;
                    q.push(v);
                    cnt[v] = cnt[u] + 1;
                    if(cnt[v] > n) return {};
                }
            }
        }
    }
    return dist;
}
```

或值最短路

```
int or_mindist(vector<tuple<int, int, int>> &edges, int n) {
    int ans = (1 << 30) - 1;
    for(int i = 29; i >= 0; i--) {
        ans ^= (1 << i);
        DSU dsu(n);
        for(auto &[u, v, w] : edges) {
            if((w & ans) == w) {
                dsu.connect(u, v);
            }
        }
        if(!dsu.is_connected(0, n - 1)) {
            ans ^= (1 << i);
        }
    }
    return ans;
}
```

Tarjan全家桶

```
// 强连通分量
struct SCC {
    explicit SCC(int n) : nodes(n), graph(n) {}
    void addedge(int u, int v) {
        if(u == v) return;
        graph[u].push_back(v);
    }
    void solve() {
        int dfn = 0, cnt = 0;
        for(int i = 0; i < nodes.size(); i++) {
            if(nodes[i].dfn == -1) {
                dfs(dfn, cnt, i);
            }
        }
        dag.assign(cnt, {});
        set<pair<int, int>> edges;
        for(int u = 0; u < graph.size(); ++u) {
            for(int v : graph[u]) {
                int bu = nodes[u].inscc, bv = nodes[v].inscc;
                if(bu != bv && !edges.contains({ bu, bv })) {
                    dag[bu].emplace_back(bv);
                    edges.insert({ bu, bv });
                }
            }
        }
    }
    struct Node {
        int dfn;
        int low;
        bool ins;
        int inscc;
        Node() : dfn(-1), low(-1), ins(false), inscc(-1) {}
    };
    vector<Node> nodes;
    vector<vector<int>> graph;
    vector<vector<int>> sccs;
    vector<vector<int>> dag;
private:
    stack<int> scc_stack;
    void dfs(int &dfn, int &cnt, int u) {
        nodes[u].dfn = dfn;
        nodes[u].low = dfn;
        ++dfn;
        scc_stack.push(u);
        nodes[u].ins = true;
        for(int v : graph[u]) {
            if(nodes[v].dfn == -1) {
                dfs(dfn, cnt, v);
                nodes[u].low = min(nodes[u].low, nodes[v].low);
            } else if(nodes[v].ins) {
                nodes[u].low = min(nodes[u].low, nodes[v].low);
            }
        }
        if(nodes[u].dfn == nodes[u].low) {
            vector<int> scc;
```

```

        int v = -1;
        while(v != u) {
            v = scc_stack.top();
            scc_stack.pop();
            nodes[v].ins = false;
            nodes[v].inscc = cnt;
            scc.emplace_back(v);
        }
        sccs.emplace_back(move(scc));
        ++cnt;
    }
}

};
// 边双
struct EBCC {
    explicit EBCC(int n) : nodes(n), graph(n), in_ebcc(n) {}
    void addedge(int u, int v) {
        if(u == v)
            return;
        graph[u].emplace_back(v);
        graph[v].emplace_back(u);
    }
    void solve() {
        int dfn = 0;
        for(int i = 0; i < nodes.size(); ++i) {
            if(nodes[i].dfn == -1) {
                dfs(i, -1, dfn);
            }
        }
        for(int i = 0; i < ebcc.size(); ++i) {
            for(int u : ebcc[i]) {
                in_ebcc[u] = i;
            }
        }
    }
    vector<vector<int>> graph;
    vector<vector<int>> ebcc;
    vector<int> in_ebcc;
private:
    struct Node {
        int dfn;
        int low;
        bool ins;
        Node() : dfn(-1), low(-1), ins(false) {}
    };
    vector<Node> nodes;
    stack<int> stk;
    void dfs(int u, int fa, int &dfn) {
        nodes[u].dfn = nodes[u].low = dfn;
        nodes[u].ins = true;
        dfn++;
        stk.push(u);
        for(int v : graph[u]) {
            if(v == fa)
                continue;
            if(nodes[v].dfn == -1) {
                dfs(v, u, dfn);
            }
        }
    }
};

```

```

        nodes[u].low = min(nodes[u].low, nodes[v].low);
    } else if(nodes[v].ins) {
        nodes[u].low = min(nodes[u].low, nodes[v].dfn);
    }
}
if(nodes[u].dfn == nodes[u].low) {
    vector<int> t;
    int n = -1;
    while(n != u) {
        n = stk.top();
        stk.pop();
        t.emplace_back(n);
        nodes[n].ins = false;
    }
    ebcc.emplace_back(move(t));
}
}
};
// 点双
struct DCC {
    explicit DCC(int n)
        : nodes(n), graph(n) {}
    void addedge(int u, int v) {
        if(u == v) return;
        graph[u].emplace_back(v);
        graph[v].emplace_back(u);
    }
    void solve() {
        int dfn_now = 0;
        for(int i = 0; i < nodes.size(); ++i) {
            if(nodes[i].dfn == -1) {
                dfs(i, -1, i, dfn_now);
            }
        }
    }
    struct Node {
        int dfn;
        int low;
        Node() : dfn(-1), low(-1) {}
    };
    vector<Node> nodes;
    vector<vector<int>> graph;
    vector<vector<int>> dcc;
private:
    stack<int> stk;
    void dfs(int u, int parent, int root, int &dfn_now) {
        nodes[u].dfn = dfn_now;
        nodes[u].low = dfn_now;
        dfn_now++;
        stk.push(u);
        int child = 0;
        for(int v : graph[u]) {
            if(v == parent) continue;
            if(nodes[v].dfn == -1) {
                dfs(v, u, root, dfn_now);
                nodes[u].low = min(nodes[u].low, nodes[v].low);
                if(nodes[v].low >= nodes[u].dfn) {

```

```

        ++child;
        vector<int> f = { u };
        while(!stk.empty()) {
            int x = stk.top();
            stk.pop();
            f.emplace_back(x);
            if(x == v) break;
        }
        dcc.emplace_back(move(f));
    }
} else {
    nodes[u].low = min(nodes[u].low, nodes[v].dfn);
}
}
if(u == root && graph[u].empty()) {
    dcc.emplace_back(vector<int>{u});
    return;
}
}
};

// 割点
struct Cutpoint {
    explicit Cutpoint(int n) : nodes(n), graph(n) {}
    void addedge(int u, int v) {
        graph[u].emplace_back(v);
        graph[v].emplace_back(u);
    }
    void solve() {
        int dfn = 0;
        for(int i = 0; i < nodes.size(); ++i) {
            if(nodes[i].dfn == -1) {
                dfs(i, -1, dfn);
            }
        }
        sort(cutpoints.begin(), cutpoints.end());
        cutpoints.erase(unique(cutpoints.begin(), cutpoints.end()), cutpoints.end());
    }
    struct Node {
        int dfn, low;
        Node() : dfn(-1), low(-1) {}
    };
    vector<Node> nodes;
    vector<vector<int>> graph;
    vector<int> cutpoints;
private:
    void dfs(int u, int fa, int &dfn) {
        nodes[u].dfn = nodes[u].low = dfn;
        ++dfn;
        int child = 0;
        bool flag = false;
        for(int v : graph[u]) {
            if(nodes[v].dfn == -1) {
                ++child;
                dfs(v, u, dfn);
                nodes[u].low = min(nodes[u].low, nodes[v].low);
                if(fa != -1) {
                    flag |= (nodes[v].low >= nodes[u].dfn);
                }
            }
        }
    }
};

```



```

        }
        } else if(v != fa) {
            nodes[u].low = min(nodes[u].low, nodes[v].dfn);
        }
    }
    if(fa == -1) {
        flag = (child > 1);
    }
    if(flag) {
        cutpoints.emplace_back(u);
    }
}

};

// 桥
struct Bridge {
    explicit Bridge(int n)
        : nodes(n), graph(n) {}
    void addedge(int u, int v) {
        graph[u].emplace_back(v);
        graph[v].emplace_back(u);
    }
    void solve() {
        int dfn_now = 0;
        for(int i = 0; i < nodes.size(); ++i) {
            if(nodes[i].dfn == -1) {
                dfs(i, -1, dfn_now);
            }
        }
    }
    struct Node {
        int dfn;
        int low;
        Node() : dfn(-1), low(-1) {}
    };
    vector<Node> nodes;
    vector<vector<int>> graph;
    vector<pair<int, int>> bridges;
private:
    void dfs(int u, int father, int &dfn_now) {
        nodes[u].dfn = nodes[u].low = dfn_now++;
        for(int v : graph[u]) {
            if(nodes[v].dfn == -1) {
                dfs(v, u, dfn_now);
                nodes[u].low = min(nodes[u].low, nodes[v].low);
                if(nodes[v].low > nodes[u].dfn) {
                    bridges.emplace_back(min(u, v), max(u, v));
                }
            } else if(v != father) {
                nodes[u].low = min(nodes[u].low, nodes[v].dfn);
            }
        }
    }
};

```

同余最短路

```
// 跑的dijkstra
ll remainder_mindist(ll k, vector<ll> dist) {
    ll modulo = 2 * min(dist[0], dist[3]);
    vector<vector<ll>> dp(4, vector<ll>(modulo, inf));
    vector<vector<bool>> visited(4, vector<bool>(modulo, false));
    dp[0][0] = 0;
    priority_queue<tuple<ll, ll, ll>, vector<tuple<ll, ll, ll>>, greater<>> pq;
    pq.emplace(0, 0, 0);
    auto get = [](ll a, ll b) -> ll {
        if(a == 3 && b == 0) {
            return 3;
        }
        if(b == 3 && a == 0) {
            return 3;
        }
        return min(a, b);
    };
    while(!pq.empty()) {
        auto [w, m, c] = pq.top();
        pq.pop();
        if(visited[c][m]) continue;
        visited[c][m] = true;
        ll d1 = (c + 1) % 4, d2 = (c + 3) % 4;
        ll w1 = dist[get(c, d1)], w2 = dist[get(c, d2)];
        if(chkmin(dp[d1][(w + w1) % modulo], w + w1)) {
            pq.emplace(w + w1, (w + w1) % modulo, d1);
        }
        if(chkmin(dp[d2][(w + w2) % modulo], w + w2)) {
            pq.emplace(w + w2, (w + w2) % modulo, d2);
        }
    }
    ll ans = inf;
    for(ll i = 0; i < modulo; ++i) {
        ll required = (k / modulo) * modulo + i;
        while(required < k) {
            required += modulo;
        }
        chkmin(ans, max(required, dp[0][i]));
    }
    return ans;
}
```

Kruskal

```
constexpr ll inf = 0x3f3f3f3f3f3f3f3f;
struct Edge {
    int u, v;
    ll w;
};
// 拿个并查集板子下来谢谢喵
ll kruskal(vector<Edge> &edges, int n) {
    DSU dsu(n);
    ll ans = 0;
    sort(edges.begin(), edges.end(), [](const Edge &a, const Edge &b) {
        return a.w < b.w;
    });
    for(auto &e : edges) {
        if(!dsu.is_connected(e.u, e.v)) {
            dsu.connect(e.u, e.v);
            ans += e.w;
        }
    }
    for(int i = 1; i < n; ++i) {
        if(!dsu.is_connected(i, 0)) {
            return inf;
        }
    }
    return ans;
}
```

Prim

```
constexpr ll inf = 0x3f3f3f3f3f3f3f3f;
ll prim(const vector<vector<pair<int, ll>>> &graph) {
    int n = graph.size();
    vector<bool> visited(n, false);
    vector<ll> dist(n, inf);
    dist[0] = 0;
    priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>> pq;
    pq.emplace(0, 0);
    ll ret = 0;
    while(!pq.empty()) {
        auto [w, v] = pq.top();
        pq.pop();
        if(visited[v]) continue;
        visited[v] = true;
        ret += w;
        for(auto [u, w] : graph[v]) {
            if(!visited[u] && dist[u] > w) {
                dist[u] = w;
                pq.emplace(w, u);
            }
        }
    }
    for(bool b : visited) {
        if(!b) {
            return inf;
        }
    }
    return ret;
}
```

Boruvka

```
constexpr ll inf = 0x3f3f3f3f3f3f3f3f;
struct Edge {
    int u, v;
    ll w;
};
ll boruvka(const vector<Edge> &edges, int n) {
    DSU dsu(n);
    ll ans = 0;
    int c = n;
    vector<Edge> mst(n);
    while(c > 1) {
        for(int i = 0; i < n; ++i) {
            if(dsu.find(i) != i) continue;
            mst[i] = Edge(-1, -1, inf);
        }
        for(auto [u, v, w] : edges) {
            int fu = dsu.find(u), fv = dsu.find(v);
            if(fu == fv) continue;
            if(mst[fu].w > w) {
                mst[fu] = Edge(u, v, w);
            }
            if(mst[fv].w > w) {
                mst[fv] = Edge(u, v, w);
            }
        }
        bool flag = false;
        for(int i = 0; i < n; ++i) {
            if(dsu.find(i) != i) continue;
            auto [u, v, w] = mst[i];
            if(w == inf) return inf;
            if(!dsu.is_connected(u, v)) {
                dsu.connect(u, v);
                ans += w;
                --c;
                flag = true;
            }
        }
        if(!flag) {
            return inf;
        }
    }
    return ans;
}
```

Dinic

```
constexpr ll inf = 0x3f3f3f3f3f3f3f3f;
struct Dinic {
    Dinic(int n, int s, int e)
        : graph(n), level(n), start(s), end(e), n(n) {}
    void addedge(int u, int v, ll w) {
        graph[u].emplace_back(Edge{ v, (int)graph[v].size(), w });
        graph[v].emplace_back(Edge{ u, (int)graph[u].size() - 1, 0 });
    }
    ll solve() {
        ll ans = 0;
        while(bfs()) {
            ll flow = dfs(start, inf);
            while(flow > 0) {
                ans += flow;
                flow = dfs(start, inf);
            }
        }
        return ans;
    }
private:
    struct Edge {
        int to;
        int rev;
        ll cap;
    };
    vector<vector<Edge>> graph;
    vector<int> level;
    int n;
    int start, end;
    bool bfs() {
        fill(level.begin(), level.end(), -1);
        level[start] = 0;
        queue<int> q;
        q.emplace(start);
        while(!q.empty()) {
            int u = q.front();
            q.pop();
            for(const auto &e : graph[u]) {
                if(level[e.to] == -1 && e.cap > 0) {
                    level[e.to] = level[u] + 1;
                    q.emplace(e.to);
                }
            }
        }
        return level[end] != -1;
    }
    ll dfs(int u, ll max_flow) {
        if(u == end || max_flow == 0) return max_flow;
        ll total_flow = 0;
        for(auto &e : graph[u]) {
            if(level[e.to] == level[u] + 1 && e.cap > 0) {
                ll min_flow = min(max_flow, e.cap);
                ll pushed = dfs(e.to, min_flow);
                if(pushed > 0) {
                    e.cap -= pushed;
                }
            }
        }
        return total_flow;
    }
};
```

```
graph[e.to][e.rev].cap += pushed;
total_flow += pushed;
max_flow -= pushed;
if(max_flow == 0) {
    return total_flow;
}
}
}
return total_flow;
}
};
```

最小费用最大流

EK

```
struct EK {
    struct Edge {
        int dst, rev;
        ll weight, flow;
        bool neg;
    };
    vector<vector<Edge>> graph;
    vector<int> pv, pe;
    vector<ll> mf, d;
    vector<bool> inq;
    int n;
    explicit EK(int _n : n(_n), graph(_n), pv(_n, -1), pe(_n, -1), d(_n, inf), mf(_n, 0), inq(_n, false) {})
    void addedge(int u, int v, ll flow, ll weight) {
        int iu = graph[u].size(), iv = graph[v].size();
        graph[u].push_back({ v, iv, weight, flow, false });
        graph[v].push_back({ u, iu, -weight, 0, true });
    }
    pair<ll, ll> solve() {
        ll cost = 0, flow = 0;
        while(true) {
            for(int i = 0; i < n; ++i) {
                d[i] = inf;
                mf[i] = 0;
                inq[i] = false;
                pv[i] = -1;
                pe[i] = -1;
            }
            queue<int> q;
            q.push(0);
            d[0] = 0;
            mf[0] = inf;
            inq[0] = true;
            while(!q.empty()) {
                int u = q.front();
                q.pop();
                inq[u] = false;
                for(int i = 0; i < graph[u].size(); ++i) {
                    auto &e = graph[u][i];
                    int v = e.dst;
                    ll f = e.flow, w = e.weight;
                    if(f > 0 && d[v] > d[u] + w) {
                        d[v] = d[u] + w;
                        pv[v] = u;
                        pe[v] = i;
                        mf[v] = min(mf[u], f);
                        if(!inq[v]) {
                            q.push(v);
                            inq[v] = true;
                        }
                    }
                }
            }
            if(mf[n - 1] <= 0) break;
        }
    }
};
```



```

    ll add = mf[n - 1];
    for(int v = n - 1; v != 0; v = pv[v]) {
        int u = pv[v], e = pe[v];
        graph[u][e].flow -= add;
        graph[v][graph[u][e].rev].flow += add;
    }
    flow += add;
    cost += add * d[n - 1];
}
return { cost, flow };
}
};

```

匈牙利算法（不带权）

```

int hungarian(const vector<vector<int>> &graph, int vsz) {
    int usz = graph.size();
    vector<int> mu(usz, -1);
    vector<int> mv(vsz, -1);
    auto dfs = [&](auto &&dfs, int u, vector<bool> &visited) -> bool {
        for(int v : graph[u]) {
            if(visited[v]) continue;
            visited[v] = true;
            if(mv[v] == -1 || dfs(dfs, mv[v], visited)) {
                mv[v] = u;
                mu[u] = v;
                return true;
            }
        }
        return false;
    };
    int ret = 0;
    for(int u = 0; u < usz; ++u) {
        if(mu[u] == -1) {
            vector<bool> visited(vsz, false);
            if(dfs(dfs, u, visited)) {
                ret++;
            }
        }
    }
    return ret;
}

```

2-SAT

```
// 本题给的是析取式，在这里我转化为了蕴含式求解
// 请打一个SCC下来
// 注意：蕴含式也要加另一条边，A->B要加B' -> A'
vector<int> solve_twosat(int n, const vector<tuple<int, int, int, int>> &conds) {
    SCC solver(2 * n);
    for(auto [i, flag_i, j, flag_j] : conds) {
        solver.addedge(2 * i + 1 - flag_i, 2 * j + flag_j);
        solver.addedge(2 * j + 1 - flag_j, 2 * i + flag_i);
    }
    solver.solve();
    for(int i = 0; i < n; ++i) {
        if(solver.nodes[2 * i].inscc == solver.nodes[2 * i + 1].inscc) {
            return {};
        }
    }
    vector<int> ans(n, -1);
    for(int i = 0; i < solver.sccs.size(); ++i) {
        for(int ii : solver.sccs[i]) {
            int u = ii / 2;
            int v = ii % 2;
            if(ans[u] == -1) {
                ans[u] = v;
            }
        }
    }
    return ans;
}
```

搜索

迭代加深

```
pair<int, stack<int>> iddfs(array<array<int, 3>, 3> src, const array<array<int, 3>, 3> &dst) {
    constexpr int dx[] = { -1, 1, 0, 0 };
    constexpr int dy[] = { 0, 0, -1, 1 };
    int ans = 0;
    stack<int> stk;
    pair<int, int> zeropos;
    for(int i = 0; i < 3; ++i) {
        for(int j = 0; j < 3; ++j) {
            if(src[i][j] == 0) {
                zeropos = { i, j };
            }
        }
    }
    auto dfs = [&](auto &&dfs, int depth, int lastdir) -> bool {
        if(src == dst) {
            for(int i = 0; i < 3; ++i) {
                for(int j = 0; j < 3; ++j) {
                    stk.push(src[i][j]);
                }
            }
            return true;
        }
        if(depth >= ans) return false;
        auto revdir = [](int cur) -> int {
            if(cur < 2) return 1 - cur;
            return 5 - cur;
        };
        for(int nowdir = 0; nowdir < 4; ++nowdir) {
            if(nowdir == revdir(lastdir)) {
                continue;
            }
            auto [oldx, oldy] = zeropos;
            int newx = oldx + dx[nowdir], newy = oldy + dy[nowdir];
            if(newx < 0 || newx > 2 || newy < 0 || newy > 2) continue;
            swap(src[oldx][oldy], src[newx][newy]);
            zeropos = { newx, newy };
            if(dfs(dfs, depth + 1, nowdir)) {
                swap(src[oldx][oldy], src[newx][newy]);
                zeropos = { oldx, oldy };
                for(int i = 0; i < 3; ++i) {
                    for(int j = 0; j < 3; ++j) {
                        stk.push(src[i][j]);
                    }
                }
                return true;
            }
            swap(src[oldx][oldy], src[newx][newy]);
            zeropos = { oldx, oldy };
        }
        return false;
    };
    for(ans = 0; ans < 33; ++ans) {
```

```
        if(dfs(dfs, 0, 114514)) {  
            return { ans, stk };  
        }  
    }  
    return { -1, stack<int>{} };  
}
```

动态规划

背包DP

```
namespace OneD {
// 01背包
ll knapsack_01(const vector<pair<int, ll>> &objects, int maxw) {
    int n = objects.size();
    vector<ll> dp(maxw + 1, 0);
    for(auto [w, v] : objects) {
        for(int j = maxw; j >= w; --j) {
            dp[j] = max(dp[j], dp[j - w] + v);
        }
    }
    return dp[maxw];
}

// 完全背包
ll knapsack_complete(const vector<pair<int, ll>> &objects, int maxw) {
    int n = objects.size();
    vector<ll> dp(maxw + 1, 0);
    for(auto [w, v] : objects) {
        for(int j = w; j <= maxw; ++j) {
            dp[j] = max(dp[j], dp[j - w] + v);
        }
    }
    return dp[maxw];
}

struct Object {
    int cost, cnt;
    ll w;
};

// 多重背包
ll multi_knapsack(const vector<Object> &objects, int maxw) {
    vector<ll> dp(maxw + 1, 0);
    for(auto [cost, cnt, w] : objects) {
        for(int _ = 0; _ < cnt; ++_) {
            for(int j = maxw; j >= cost; --j) {
                dp[j] = max(dp[j], dp[j - cost] + w);
            }
        }
    }
    return dp[maxw];
}

// 多重背包的二进制优化
ll multi_knapsack_binary(const vector<Object> &objects, int maxw) {
    vector<pair<int, ll>> objs;
    for(auto [cost, cnt, w] : objects) {
        int k = 1;
        while(cnt > 0) {
            int take = min(k, cnt);
            objs.emplace_back(cost * take, w * take);
            cnt -= take;
            k *= 2;
        }
    }
    return knapsack_01(objs, maxw);
}
```

```
}
// 缺少：多重背包的单调队列优化
}

namespace TwoD {
struct Object {
    int u, v;
    ll w;
};
// 二维01背包
ll knapsack_01(int U, int V, const vector<Object> &objects) {
    vector<vector<ll>> dp(U + 1, vector<ll>(V + 1, 0));
    for(auto [u, v, w] : objects) {
        for(int i = U; i - u >= 0; --i) {
            for(int j = V; j - v >= 0; --j) {
                dp[i][j] = max(dp[i][j], dp[i - u][j - v] + w);
            }
        }
    }
    return dp[U][V];
}
}
```

区间DP

```
int main() {
    cin.tie(nullptr)->sync_with_stdio(false);
    int n;
    cin >> n;
    vector<ll> nums(n);
    for(int i = 0; i < n; ++i) {
        cin >> nums[i];
    }
    vector<int> s(2 * n + 1);
    for(int i = 0; i < 2 * n; ++i) {
        s[i + 1] = s[i] + nums[i];
    }
    vector<vector<int>> dp(2 * n + 1, vector<int>(2 * n + 1));
    for(int len = 2; len <= n; ++len) {
        for(int i = 0; i <= 2 * n - len; ++i) {
            int j = i + len;
            int mx = 0;
            for(int k = i + 1; k < j; ++k) {
                mx = max(mx, dp[i][k] + dp[k][j]);
            }
            dp[i][j] = mx + s[j] - s[i];
        }
    }
    int ans = 0;
    for(int i = 0; i < n; ++i) {
        ans = max(ans, dp[i][i + n]);
    }
    cout << ans << '\n';
    return 0;
}
```

数位DP

```
constexpr ll modulo = 10'000'0007;
ll digit_dp(string k, int d) {
    vector<vector<ll>> memo(k.size(), vector<ll>(d, -1));
    auto dfs = [&](auto &&dfs, int idx, ll psum, bool isnum, bool islimit) -> ll {
        if(idx == k.size()) {
            return (ll)(isnum && (psum == 0));
        }
        if(memo[idx][psum] != -1 && isnum && !islimit) {
            return memo[idx][psum];
        }
        ll ans = 0;
        if(!isnum) {
            ans = (ans + dfs(dfs, idx + 1, 0, false, false)) % modulo;
        }
        int llim = isnum ? 0 : 1, hlim = islimit ? (k[idx] - '0') : 9;
        for(int i = llim; i <= hlim; ++i) {
            ans = (ans + dfs(dfs, idx + 1, ((psum + i) % d), true, (islimit && (i == hlim)))) % modulo;
        }
        if(isnum && !islimit) {
            memo[idx][psum] = ans;
        }
        return ans;
    };
    return dfs(dfs, 0, 0, false, true);
}
```


计数DP

```
constexpr ll modulo = 10'000'0007;
ll count_dp(string s) {
    int n = s.size();
    vector<vector<ll>> dp(n, vector<ll>(n, 0));
    dp[0][0] = 1;
    vector<vector<ll>> sum(n, vector<ll>(n + 1, 0));
    sum[0][1] = 1;
    for(int i = 1; i < n; ++i) {
        for(int j = 0; j <= i; ++j) {
            if(s[i - 1] == '<') {
                dp[i][j] = sum[i - 1][j];
            } else {
                dp[i][j] = (sum[i - 1][i] - sum[i - 1][j] + modulo) % modulo;
            }
        }
        sum[i][0] = 0;
        for(int j = 1; j <= i + 1; ++j) {
            sum[i][j] = (sum[i][j - 1] + dp[i][j - 1]) % modulo;
        }
    }
    ll ans = 0;
    for(int j = 0; j < n; ++j) {
        ans = (ans + dp[n - 1][j]) % modulo;
    }
}
```

线段树优化DP

```
// SegTree区间最大值、单点加法线段树
int lis(const vector<int> &nums) {
    auto discrete = nums;
    sort(discrete.begin(), discrete.end());
    discrete.erase(unique(discrete.begin(), discrete.end()), discrete.end());
    unordered_map<int, int> mp;
    for(int i = 0; i < discrete.size(); i++) {
        mp[discrete[i]] = i;
    }
    SegTree seg(discrete.size());
    for(int i = 0; i < nums.size(); i++) {
        int x = mp[nums[i]];
        int v = seg.query(0, x) + 1;
        seg.update(x, v);
    }
    return seg.query(0, discrete.size());
}
```

数学

快速幂和模整数类

```
ll qpow(ll x, ll n) {
    ll ret = 1;
    for(; n != 0; n >>= 1, x = x * x % modulo) {
        if(n & 1) ret = ret * x % modulo;
    }
    return ret;
}

struct PreprocessedPow {
    PreprocessedPow(ll k, ll maxn) {
        k %= modulo;
        m = ceil(sqrt(maxn + 1.5));
        powerk.assign(m, 1);
        powerkm.assign(m, 1);
        for(int i = 1; i < m; ++i) {
            powerk[i] = (ll(powerk[i - 1]) * k) % modulo;
        }
        powerkm[1] = ll(powerk[m - 1]) * k % modulo;
        for(int i = 2; i < m; ++i) {
            powerkm[i] = ll(powerkm[i - 1]) * ll(powerkm[1]) % modulo;
        }
    }
    ll pow(ll n) const {
        ll i = n / m, j = n % m;
        return (ll(powerkm[i]) * ll(powerk[j])) % modulo;
    }
private:
    int m;
    vector<int> powerk, powerkm;
};

struct ModInt {
    ModInt(ll v = 0) : val(v % modulo) {
        if(val < 0) val += modulo;
    }
    ModInt operator+(const ModInt &rhs) const {
        ll retval = val + rhs.val;
        if(retval >= modulo) retval -= modulo;
        return ModInt(retval, true);
    }
    ModInt operator-(const ModInt &rhs) const {
        ll retval = val - rhs.val;
        if(retval < 0) retval += modulo;
        return ModInt(retval, true);
    }
    ModInt operator*(const ModInt &rhs) const {
        return ModInt(val * rhs.val % modulo, true);
    }
    ModInt operator/(const ModInt &rhs) const {
        return ModInt(val * qpow(rhs.val, modulo - 2) % modulo, true);
    }
    ModInt power(int n) const {
```

```

        ModInt ret = 1;
        for (ModInt base = val; n != 0; n >>= 1, base = base * base) {
            if (n & 1) ret = ret * base;
        }
        return ret;
    }
    operator ll() const {
        return val;
    }
private:
    ll val;
    ModInt(ll v, bool) : val(v) {}
};

```

组合数

```

struct Comb {
    Comb() = delete;
    static ll fact(ll n) {
        return _fact[n];
    }
    static ll invfact(ll n) {
        return _invfact[n];
    }
    static ll binom(ll n, ll m) {
        if (m < 0 || m > n) return 0;
        return (((ll(_fact[n]) * ll(_invfact[m])) % modulo) * ll(_invfact[n - m])) % modulo;
    }
private:
    static constexpr int _maxn = 500005;
    static inline int _fact[_maxn], _invfact[_maxn];
    static inline int init = [&] {
        _fact[0] = 1;
        for (ll i = 1; i < _maxn; ++i) {
            _fact[i] = (ll(_fact[i - 1]) * i) % modulo;
        }
        _invfact[_maxn - 1] = qpow(_fact[_maxn - 1], modulo - 2);
        for (ll i = _maxn - 2; i >= 0; --i) {
            _invfact[i] = (ll(_invfact[i + 1]) * (i + 1)) % modulo;
        }
        return 0;
    }();
};

```

卡特兰数

```
struct Catalan {
    Catalan() = delete;
    static ll get(int x) {
        return _cat[x];
    }
private:
    static constexpr int _maxn = 250005;
    static inline int _cat[_maxn];
    static inline int init = [] {
        _cat[0] = 1;
        for(int i = 1; i < _maxn; ++i) {
            _cat[i] = (Comb::binom(2 * i, i) - Comb::binom(2 * i, i - 1) + modulo) % modulo;
        }
        return 0;
    }();
};
```

EXGCD

```
// @returns (gcd, x, y) so that gcd = ax + by
tuple<ll, ll, ll> exgcd(ll a, ll b) {
    if(b == 0) return tuple(a, 1ll, 0ll);
    auto [g, x, y] = exgcd(b, a % b);
    return tuple(g, y, x - (a / b) * y);
}
```

中国剩余定理

```
// @returns (a, b) so that answer is a + kb, k\in N_+
pair<ll, ll> crt(const vector<ll> &rem, const vector<ll> &mod) {
    int n = rem.size();
    ll modulo_ = 1, ans = 0;
    for(int i = 0; i < n; ++i) {
        modulo_ *= mod[i];
    }
    for(int i = 0; i < n; ++i) {
        ll m = modulo_ / mod[i];
        auto [_, b, __] = exgcd(m, mod[i]);
        ans = (ans + ((rem[i] * m) % modulo_ * b) % modulo_) % modulo_;
    }
    return pair((ans % modulo_ + modulo_) % modulo_, modulo_);
}

pair<ll, ll> excrt(const vector<ll> &rem, const vector<ll> &mod) {
    int n = rem.size();
    ll r1 = rem[0], m1 = mod[0];
    for(int i = 1; i < n; ++i) {
        ll r2 = rem[i], m2 = mod[i];
        auto [g, p, _] = exgcd(m1, m2);
        if((r2 - r1) % g != 0) {
            return { -1, -1 };
        }
        ll v = m2 / g, x = (r2 - r1) / g;
        ll u = p % v * x % v;
        ll w = (u % v + v) % v;
        r1 += w * m1;
        m1 = lcm(m1, m2);
    }
    return { (r1 % m1 + m1) % m1, m1 };
}
```

整除分块

```
ll intdiv(ll n) {
    ll ans = 0;
    for(ll b = 1; b <= n; ) {
        ll val = n / b;
        ll r = n / val;
        ll len = r - b + 1;
        ans = (ans + (len % modulo * val % modulo)) % modulo;
        b = r + 1;
    }
    return ans;
}
```

二项式反演公式

$$f(n) = \sum_{k=0}^n \binom{n}{k} g(k) \rightarrow g(n) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(k)$$

$$f(n) = \sum_{k=n}^m \binom{k}{n} g(k) \rightarrow g(n) = \sum_{k=n}^m (-1)^{k-n} \binom{k}{n} f(k)$$

斯特林数

预处理

```
struct Sterling {
    Sterling() = delete;
    static int get(int n, int m) {
        return ster[n][m];
    }
private:
    static constexpr int maxn = 5005;
    static inline int ster[maxn][maxn];
    static inline int init = [&] {
        ster[0][0] = 1;
        for(ll i = 1; i < maxn; ++i) {
            ster[i][0] = 0;
            for(ll j = 1; j < i; ++j) {
                ster[i][j] = (1ll(ster[i - 1][j - 1]) + 1ll(ster[i - 1][j]) * j) % modulo;
            }
        }
        return 0;
    }();
};
```

使用NTT求一行

得到的是 $\left\{ \begin{smallmatrix} 0 \\ k \end{smallmatrix} \right\}, \left\{ \begin{smallmatrix} 1 \\ k \end{smallmatrix} \right\}, \dots, \left\{ \begin{smallmatrix} k \\ k \end{smallmatrix} \right\}$

```

vector<ll> sterling_ntt(int k) {
    if(k == 0) return {1};
    vector<ll> powk(k + 1, 0);
    powk[1] = 1;
    vector<bool> isprime(k + 1, true);
    isprime[0] = isprime[1] = false;
    vector<int> primes;
    for(int i = 2; i <= k; ++i) {
        if(isprime[i]) {
            primes.push_back(i);
            powk[i] = qpow(i, k);
        }
        for(ll p : primes) {
            ll q = i * p;
            if(q > k) break;
            isprime[q] = false;
            powk[q] = powk[i] * powk[p] % modulo;
            if(i % p == 0) break;
        }
    }
    vector<ll> fact(k + 1), invfact(k + 1);
    fact[0] = fact[1] = 1;
    for(ll i = 2; i <= k; ++i) {
        fact[i] = fact[i - 1] * i % modulo;
    }
    invfact[k] = qpow(fact[k], modulo - 2);
    for(int i = k - 1; i >= 0; --i) {
        invfact[i] = invfact[i + 1] * (i + 1) % modulo;
    }
    int n = 1;
    while(n < (2 * k + 1)) {
        n <<= 1;
    }
    vector<ll> a(n, 0), b(n, 0);
    for(int i = 0; i <= k; ++i) {
        a[i] = powk[i] * invfact[i] % modulo;
        if(i % 2 == 1) {
            b[i] = (modulo - invfact[i]) % modulo;
        } else {
            b[i] = invfact[i];
        }
    }
}

auto ntt = [](auto &&ntt, vector<ll> &f, bool invert) -> void {
    int n = f.size();
    if(n == 1) return;
    vector<ll> f0(n / 2), f1(n / 2);
    for(int i = 0; i < n / 2; ++i) {
        f0[i] = f[2 * i];
        f1[i] = f[2 * i + 1];
    }
    ntt(ntt, f0, invert);
    ntt(ntt, f1, invert);
    ll w = 1, wn = qpow(g, (modulo - 1) / n);
    if(invert) {
        wn = qpow(wn, modulo - 2);
    }
    for(int i = 0; i < n / 2; ++i) {

```

```

        ll u = f0[i];
        ll v = w * f1[i] % modulo;
        f[i] = (u + v) % modulo;
        f[i + n / 2] = (u - v + modulo) % modulo;
        w = w * wn % modulo;
    }
};

ntt(ntt, a, false);
ntt(ntt, b, false);
vector<ll> c(n);
for(int i = 0; i < n; ++i) {
    c[i] = a[i] * b[i] % modulo;
}
ntt(ntt, c, true);
ll invn = qpow(n, modulo - 2);
for(int i = 0; i < n; ++i) {
    c[i] = c[i] * invn % modulo;
}
c.resize(k + 1);
return c;
}

```


质数筛和它能干的事

```
ll __qpow(ll x, ll n) {
    ll ret = 1;
    for(; n != 0; n >>= 1, x = x * x) {
        if(n & 1) ret = ret * x;
    }
    return ret;
}

template<class F> concept eulersieve_func = convertible_to<F, function<ll(int, int)>>;

struct EulerSieve {
    vector<int> primes, lpf, lpow;
    vector<ll> fv;

    explicit EulerSieve(int n) : lpf(n + 1), lpow(n + 1) {
        for(ll i = 2; i <= n; ++i) {
            if(lpf[i] == 0) {
                lpf[i] = i;
                lpow[i] = 1;
                primes.push_back(i);
            }
            for(ll p : primes) {
                ll j = i * p;
                if(j > n) break;
                lpf[j] = p;
                if(i % p == 0) {
                    lpow[j] = lpow[i] + 1;
                    ll jp = __qpow(lpf[j], lpow[j]);
                    ll rem = j / jp;
                    break;
                } else {
                    lpow[j] = 1;
                }
            }
        }
    }

    template<eulersieve_func F> EulerSieve(int n, F &&f) : lpf(n + 1), lpow(n + 1), fv(n + 1) {
        fv[1] = 1;
        for(ll i = 2; i <= n; ++i) {
            if(lpf[i] == 0) {
                lpf[i] = i;
                lpow[i] = 1;
                primes.push_back(i);
                fv[i] = f(i, 1);
            }
            for(ll p : primes) {
                ll j = i * p;
                if(j > n) break;
                lpf[j] = p;
                if(i % p == 0) {
                    lpow[j] = lpow[i] + 1;
                    ll jp = __qpow(lpf[j], lpow[j]);
                    ll rem = j / jp;
                    if(rem == 1) {
```

```

        fv[j] = f(p, lpow[j]);
    } else {
        fv[j] = fv[rem] * fv[jp];
    }
    break;
} else {
    fv[j] = fv[i] * fv[p];
    lpow[j] = 1;
}
}
}
}

};

int euler_f(int p, int k) {
    return __qpow(p, k) - __qpow(p, k - 1);
}

int mobius_f(int p, int k) {
    return k == 0 ? 1 : k == 1 ? -1 : 0;
}

int factor_cnt_f(int p, int k) {
    return k + 1;
}

int factor_sum_f(int p, int k) {
    return (__qpow(p, k + 1) - 1) / (p - 1);
}

```

莫比乌斯反演公式

$$F(n) = \sum_{d|n} f(d) \leftrightarrow f(n) = \sum_{d|n} \mu(d) F\left(\frac{n}{d}\right)$$

$$f * 1 = F \leftrightarrow f = F * \mu$$

多项式乘法

FFT

```
using ld = long double;
using cd = complex<ld>;
constexpr ld pi = 3.141592653589793238462643383279502881;
vector<ld> multiply(const vector<ld> &a, const vector<ld> &b) {
    int n = 1;
    while(n < (a.size() + b.size())) {
        n <<= 1;
    }
    vector<cd> fa(n), fb(n);
    for(int i = 0; i < a.size(); ++i) {
        fa[i] = a[i];
    }
    for(int i = 0; i < b.size(); ++i) {
        fb[i] = b[i];
    }
    auto fft = [](auto &&fft, vector<cd> &f, bool invert) -> void {
        int n = f.size();
        if(n == 1) return;
        vector<cd> f0(n / 2), f1(n / 2);
        for (int i = 0; i < n / 2; ++i) {
            f0[i] = f[2 * i];
            f1[i] = f[2 * i + 1];
        }
        fft(fft, f0, invert);
        fft(fft, f1, invert);
        ld theta = 2.1 * pi / n * (invert ? -1.1 : 1.1);
        cd wt = 1, w(cos(theta), sin(theta));
        for (int t = 0; t < n / 2; ++t) {
            cd u = f0[t], v = wt * f1[t];
            f[t] = u + v;
            f[t + n / 2] = u - v;
            wt *= w;
        }
    };
    fft(fft, fa, false);
    fft(fft, fb, false);
    for(int i = 0; i < n; ++i) {
        fa[i] *= fb[i];
    }
    fft(fft, fa, true);
    for(int i = 0; i < n; ++i) {
        fa[i] /= n;
    }
    vector<ld> ret(n);
    for(int i = 0; i < n; ++i) {
        ret[i] = fa[i].real();
    }
    while(ret.size() > (a.size() + b.size() - 1)) {
        ret.pop_back();
    }
    return ret;
}
```

NTT

```
constexpr ll modulo = 9'9824'4353, g = 3;
// constexpr ll modulo = 10'0000'0007, g = 5; 不可以

ll qpow(ll x, ll n) {
    ll ret = 1;
    while(n) {
        if(n & 1) {
            ret = ret * x % modulo;
        }
        x = x * x % modulo;
        n >>= 1;
    }
    return ret;
}

vector<ll> multiply(const vector<ll> &a, const vector<ll> &b) {
    int n = 1;
    while(n < a.size() + b.size()) {
        n <<= 1;
    }
    vector<ll> ca(n), cb(n);
    for(int i = 0; i < a.size(); ++i) {
        ca[i] = a[i];
    }
    for(int i = 0; i < b.size(); ++i) {
        cb[i] = b[i];
    }
    auto ntt = [](auto &&ntt, vector<ll> &f, bool invert) -> void {
        int n = f.size();
        if(n == 1) return;
        vector<ll> f0(n / 2), f1(n / 2);
        for(int i = 0; i < n / 2; ++i) {
            f0[i] = f[2 * i];
            f1[i] = f[2 * i + 1];
        }
        ntt(ntt, f0, invert);
        ntt(ntt, f1, invert);
        ll w = 1, wn = qpow(g, (modulo - 1) / n);
        if(invert) {
            wn = qpow(wn, modulo - 2);
        }
        for(int t = 0; t < n / 2; ++t) {
            ll u = f0[t], v = w * f1[t] % modulo;
            f[t] = (u + v) % modulo;
            f[t + n / 2] = (u - v + modulo) % modulo;
            w = w * wn % modulo;
        }
    };
    ntt(ntt, ca, false);
    ntt(ntt, cb, false);
    for(int i = 0; i < n; ++i) {
        ca[i] = ca[i] * cb[i] % modulo;
    }
    ntt(ntt, ca, true);
    for(int i = 0; i < n; ++i) {
        ca[i] = ca[i] * qpow(n, modulo - 2) % modulo;
    }
}
```

```
while(ca.size() > (a.size() + b.size() - 1)) {  
    ca.pop_back();  
}  
return ca;  
}
```

FWT

```
ll modadd(ll a, ll b) {
    a += b;
    if(a >= modulo) a -= modulo;
    return a;
}

ll modsub(ll a, ll b) {
    a -= b;
    if(a < 0) a += modulo;
    return a;
}

void fwt_and(vector<ll> &nums, bool invert) {
    int n = nums.size();
    for(ll x = 2; x <= n; x <= 1) {
        ll k = x >> 1;
        for (ll i = 0; i < n; i += x) {
            for (ll j = 0; j < k; j++) {
                if(invert) {
                    nums[i + j] = modsub(nums[i + j], nums[i + j + k]);
                } else {
                    nums[i + j] = modadd(nums[i + j], nums[i + j + k]);
                }
            }
        }
    }
}

void fwt_or(vector<ll> &nums, bool invert) {
    int n = nums.size();
    for (ll x = 2; x <= n; x <= 1) {
        ll k = x >> 1;
        for (ll i = 0; i < n; i += x) {
            for (ll j = 0; j < k; j++) {
                if(invert) {
                    nums[i + j + k] = modsub(nums[i + j + k], nums[i + j]);
                } else {
                    nums[i + j + k] = modadd(nums[i + j + k], nums[i + j]);
                }
            }
        }
    }
}

void fwt_xor(vector<ll> &nums, bool invert) {
    int n = nums.size();
    for (ll x = 2; x <= n; x <= 1) {
        ll k = x >> 1;
        for (ll i = 0; i < n; i += x) {
            for (ll j = 0; j < k; j++) {
                nums[i + j] = modadd(nums[i + j], nums[i + j + k]);
                nums[i + j + k] = modsub(nums[i + j], modadd(nums[i + j + k], nums[i + j + k]));
                if(invert) {
                    nums[i + j] = modulo - nums[i + j];
                    nums[i + j + k] = modulo - nums[i + j + k];
                }
            }
        }
    }
}
```

```

    }
}
}
}

```

线性基

```

struct LinearBasis_XOR {
    LinearBasis_XOR() : base(61) {}
    void insert(ll val) {
        for(int i = 60; i >= 0; --i) {
            if(((val >> i) & 1) == 0) continue;
            if(base[i] == 0) {
                base[i] = val;
                return;
            }
            val ^= base[i];
        }
    }
    ll query_max() const {
        ll ans = 0;
        for(int i = 60; i >= 0; --i) {
            if((ans ^ base[i]) > ans) {
                ans ^= base[i];
            }
        }
        return ans;
    }
private:
    vector<ll> base;
};

```

矩阵乘法 and 快速幂

```
template<class T> vector<vector<T>> matmul(const vector<vector<T>> &lhs, const vector<vector<T>> &rhs) {
    int M = lhs.size(), N = lhs[0].size(), P = rhs[0].size();
    vector<vector<T>> ret(M, vector<T>(P, 0));
    for(int i = 0; i < M; ++i) {
        for(int j = 0; j < P; ++j) {
            for(int k = 0; k < N; ++k) {
                ret[i][j] = ret[i][j] + lhs[i][k] * rhs[k][j];
            }
        }
    }
    return ret;
}

template<class T> vector<vector<T>> matpow(vector<vector<T>> mat, ll N) {
    int M = mat.size();
    vector<vector<T>> ret(M, vector<T>(M, 0));
    for(int i = 0; i < M; ++i) {
        ret[i][i] = static_cast<T>(1);
    }
    while(N) {
        if(N & 1) {
            ret = matmul(ret, mat);
        }
        mat = matmul(mat, mat);
        N >>= 1;
    }
    return ret;
}
```


高斯消元

```
vector<vector<ll>> gauss(const vector<vector<ll>> &a, const vector<vector<ll>> &b) {
    int r = a.size(), n = a[0].size(), m = b[0].size(), row = 0;
    vector<vector<ll>> aug(r, vector<ll>(n + m));
    for(int i = 0; i < r; ++i) {
        for(int j = 0; j < m + n; ++j) {
            aug[i][j] = (j < n) ? a[i][j] : b[i][j - n];
        }
    }
    vector<int> where(n, -1);
    for(int col = 0; col < n && row < r; ++col) {
        int sel = row;
        while(sel < r && aug[sel][col] == 0) ++sel;
        if(sel == r) continue;
        swap(aug[row], aug[sel]);
        ll inv = qpow(aug[row][col], modulo - 2);
        for(int j = col; j < n + m; ++j) {
            aug[row][j] = aug[row][j] * inv % modulo;
        }
        for(int i = 0; i < r; ++i) {
            if(i != row && aug[i][col]) {
                ll f = aug[i][col];
                for(int j = col; j < n + m; ++j) {
                    aug[i][j] = (aug[i][j] - f * aug[row][j] % modulo + modulo) % modulo;
                }
            }
        }
        where[col] = row;
        ++row;
    }
    for(int i = row; i < r; ++i) {
        bool flag = true;
        for(int j = 0; j < n; ++j) {
            if(aug[i][j] != 0) {
                flag = false;
                break;
            }
        }
        if(flag) {
            for(int j = 0; j < m; ++j) {
                if(aug[i][n + j]) throw runtime_error("No solution");
            }
        }
    }
    for(int col = 0; col < n; ++col) {
        if(where[col] == -1) throw runtime_error("Multiple solutions");
    }
    vector<vector<ll>> ret(n, vector<ll>(m, 0));
    for(int col = 0; col < n; ++col) {
        for(int j = 0; j < m; ++j) {
            ret[col][j] = aug[where[col]][n + j];
        }
    }
    return ret;
}
```

构造增广状态转移矩阵

问题	转移矩阵	初始值	结果
$\sum_{k=0}^n (aB^k)_{i,j}$	$T = \begin{bmatrix} B & 0 \\ e_j \cdot e_i^T & 1 \end{bmatrix}$	$z_0 = \begin{bmatrix} a \\ a_i \end{bmatrix}$	$z_n = T^n z_0$ 的 最后一个分量
$\sum_{k=0}^n \langle u, B^k v \rangle$	$T = \begin{bmatrix} B & 0 \\ u^T & 1 \end{bmatrix}$	$z_0 = \begin{bmatrix} v \\ \langle u, v \rangle \end{bmatrix}$	$s_n = T^n z_0[back]$
$\sum_{i=0}^n A^i$	$T = \begin{bmatrix} A & I \\ 0 & I \end{bmatrix}$	无	T^{n+1} 的 右上角分块

Lehmer码

`l[i] = k` 代表后面有k个数比 `a[i]` 小

```
// 均为0-based排列
vector<int> lehmer(const vector<int> &a) {
    int n = a.size();
    vector<int> l(n);
    Fenwick bit(n);
    for(int i = 1; i <= n; ++i) {
        bit.update(i, 1);
    }
    for(int i = 0; i < n; ++i) {
        int x = a[i] + 1;
        l[i] = bit.query(n) - bit.query(x);
        bit.update(x, -1);
    }
    return l;
}

vector<int> rev_lehmer(const vector<int> &l) {
    int n = l.size();
    VST vst(n); // 使用权值线段树实现会快一点
    for(int i = 0; i < n; ++i) {
        vst.insert(i);
    }
    vector<int> ret(n);
    for(int i = 0; i < n; ++i) {
        ret[i] = vst.qvr(l[i] + 1);
        vst.erase(ret[i]);
    }
    return ret;
}
```

计算几何

常量与数据类型

```
using ld = long double;
constexpr ld eps = 1e-9;
int sign(ld a) {
    return (a < -eps) ? -1 : (a > eps) ? 1 : 0;
}
int cmp(ld a, ld b) {
    return sign(a - b);
}
constexpr ld pi = 3.14159265358979323846261;
constexpr ld inf = 1e121;
```

点和向量

```
struct Point {
    ld x, y;
    Point() : x(0.01), y(0.01) {}
    Point(ld _x, ld _y) : x(_x), y(_y) {}
    Point(const complex<ld> &cd) : x(cd.real()), y(cd.imag()) {}
    operator complex<ld>() const {
        return complex<ld>(x, y);
    }
    Point operator+(const Point &p) const {
        return Point(x + p.x, y + p.y);
    }
    Point operator-(const Point &p) const {
        return Point(x - p.x, y - p.y);
    }
    Point operator*(ld z) const {
        return Point(z * x, z * y);
    }
    friend Point operator*(ld z, const Point &p) {
        return p * z;
    }
    Point operator/(ld z) const {
        return Point(x / z, y / z);
    }
    bool operator==(const Point &p) const {
        return cmp(x, p.x) == 0 && cmp(y, p.y) == 0;
    }
    ld len() const {
        return sqrt(x * x + y * y);
    }
    // [0, 2pi)
    ld arg() const {
        ld ret = atan2(y, x);
        int c = cmp(ret, 0);
        return c == 1 ? ret : c == 0 ? 0.01 : ret + 2 * pi;
    }
    Point rotate(ld a) const {
        return Point(x * cos(a) - y * sin(a), x * sin(a) + y * cos(a));
    }
};

using Vector = Point;
ld dot(const Vector &x, const Vector &y) {
    return x.x * y.x + x.y * y.y;
}
ld cross(const Vector &x, const Vector &y) {
    return x.x * y.y - x.y * y.x;
}
ld cross(const Point &o, const Point &a, const Point &b) {
    return cross(a - o, b - o);
}
bool argcmp(const Point &x, const Point &y) {
    bool bx = sign(x.y) == 1 || (sign(x.y) == 0 && sign(x.x) == 1),
        by = sign(y.y) == 1 || (sign(y.y) == 0 && sign(y.x) == 1);
    if(bx != by) return bx;
    return sign(cross(x, y)) == 0;
}
```

```

ld dist(const Point &x, const Point &y) {
    return (x - y).len();
}
int to_left(const Vector &a, const Vector &b) {
    return sign(cross(a, b));
}
int to_left(const Point &a, const Point &b, const Point &c) {
    return sign(cross(a, b, c));
}

```

直线

```

struct Line {
    Point p, v;
    Line() {}
    Line(const Point &p, const Vector &v) : p(p), v(v) {}
};
int to_left(const Line &ln, const Point &p) {
    return to_left(ln.v, p - ln.p);
}
bool parallel(const Line &l1, const Line &l2) {
    return cmp(cross(l1.v, l2.v), 0.01) == 0;
}
int is_inter(const Line &l1, const Line &l2) {
    return parallel(l1, l2) ? 0 : 1;
}
Point inter(const Line &a, const Line &b) {
    if(parallel(a, b))
        throw invalid_argument("a and b are parallel");
    Vector v = a.v * (cross(b.v, a.p - b.p) / cross(a.v, b.v));
    return a.p + v;
}
ld dist(const Point &p, const Line &ln) {
    return abs(cross(ln.v, p - ln.p)) / ln.v.len();
}
ld dist(const Line &l1, const Line &l2) {
    if(!parallel(l1, l2)) {
        return 0;
    }
    return dist(l1.p, l2);
}
Point proj(const Point &p, const Line &ln) {
    Vector v = ln.v * (dot(ln.v, p - ln.p) / (dot(ln.v, ln.v)));
    return ln.p + v;
}
bool is_on(const Point &p, const Line &ln) {
    return cmp(cross(ln.v, ln.p - p), 0.01) == 0;
}

```

线段

```
struct Lineseg {
    Point a, b;
    Lineseg() {}
    Lineseg(const Point &a, const Point &b) : a(_a), b(_b) {}
    ld len() const {
        return (b - a).len();
    }
};

int is_on(const Point &p, const Lineseg &ls) {
    if(p == ls.a || p == ls.b) {
        return 2;
    }
    return to_left(p - ls.a, p - ls.b) == 0 && cmp(dot(p - ls.a, p - ls.b), 0) == -1;
}

int is_inter(const Line &ln, const Lineseg &ls) {
    int a = to_left(ln, ls.a), b = to_left(ln, ls.b);
    if(a == 0 || b == 0) {
        return 2;
    }
    return a == b ? 0 : 1;
}

int is_inter(const Lineseg &l1, const Lineseg &l2) {
    if(is_on(l1.a, l2) || is_on(l1.b, l2) || is_on(l2.a, l1) || is_on(l2.b, l1)) {
        return 2;
    }
    Line ln1(l1.a, l1.b - l1.a), ln2(l2.a, l2.b - l2.a);
    return to_left(ln1, l2.a) * to_left(ln1, l2.b) == -1 && to_left(ln2, l1.a) * to_left(ln2, l1.b) == -1;
}

ld dist(const Point &p, const Lineseg &ls) {
    if(is_on(p, ls) != 0) {
        return 0.01;
    }
    if(cmp(dot(p - ls.a, ls.b - ls.a), 0) == -1 || cmp(dot(p - ls.b, ls.a - ls.b), 0) == -1) {
        return min(dist(p, ls.a), dist(p, ls.b));
    }
    Line l(ls.a, ls.b - ls.a);
    return dist(p, l);
}
```

多边形

```
struct Polygon {
    vector<Point> pts;
    Polygon() {}
    Polygon(const vector<Point> &p) : pts(p) {}
    Polygon(const vector<Line> &l) {
        pts.reserve(l.size());
        for(int i = 0; i < l.size(); ++i) {
            pts.emplace_back(inter(l[i], l[(i + 1) % l.size()]));
        }
    }
    ld area() const {
        ld ret = 0.01;
        for(int i = 0; i < pts.size(); ++i) {
            ret += cross(pts[i], pts[(i + 1) % pts.size()]);
        }
        return ret / 2.01;
    }
    ld circ() const {
        ld ret = 0.01;
        for(int i = 0; i < pts.size(); ++i) {
            ret += (pts[i] - pts[(i + 1) % pts.size()]).len();
        }
        return ret;
    }
};
```



```
struct Circle {
    Point c;
    ld r;
    Circle() : r(0.01) {}
    Circle(const Point &c, ld _r) : c(_c), r(_r) {}
    ld area() const {
        return pi * r * r;
    }
    ld circ() const {
        return 2.01 * pi * r;
    }
};

vector<Point> inter(const Circle &c1, const Circle &c2) {
    ld dis = (c2.c - c1.c).len();
    ld r1 = c1.r, r2 = c2.r;
    if(abs(dis) <= eps) {
        if(r1 == r2) {
            return {Point{}};
        }
        return {};
    }
    if (dis > r1 + r2 + eps || dis < abs(r1 - r2) + eps) {
        return {};
    }
    ld cosa = ((r1 * r1 + dis * dis - r2 * r2) / (2 * r1 * dis));
    ld alp = acos(min(max(cosa, -1.01), 1.01));
    Point v = c2.c - c1.c;
    v = v / v.len() * c1.r;
    return {c1.c + v.rotate(alp), c1.c + v.rotate(-alp)};
}

bool inter(const Circle &c, const Line &l, pair<Point, Point> &ret) {
    ld d = dist(c.c, l);
    if(sign(d) == 0) {
        ret.first = l.v * (1.01 / l.v.len()) * c.r + c.c;
        ret.second = l.v * (-1.01 / l.v.len()) * c.r + c.c;
        return true;
    }
    if(cmp(c.r, d) != 1) return false;
    Point vec;
    if(to_left(l.v, c.c - l.p) == 1) {
        vec = Point{ l.v.y, -l.v.x };
    } else {
        vec = Point{ -l.v.y, l.v.x };
    }
    vec = vec * (1.01 / vec.len()) * c.r;
    ld cosa = d / c.r;
    ld a = acos(cosa);
    ret.first = vec.rotate(-a) + c.c;
    ret.second = vec.rotate(a) + c.c;
    return true;
}
```


平面凸包

```
vector<Point> andrew(vector<Point> points) {
    sort(points.begin(), points.end(), [](const Point &a, const Point &b) {
        return a.x < b.x || (a.x == b.x && a.y < b.y);
    });
    points.erase(unique(points.begin(), points.end()), points.end());
    int n = points.size();
    if(n <= 2) {
        return points;
    }
    vector<Point> stk;
    for(int i = 0; i < n; ++i) {
        while(stk.size() >= 2 && sign(cross(stk[stk.size() - 2], stk.back(), points[i])) <= 0) {
            stk.pop_back();
        }
        stk.push_back(points[i]);
    }
    int t = stk.size();
    for(int i = n - 2; i >= 0; --i) {
        while(stk.size() > t && sign(cross(stk[stk.size() - 2], stk.back(), points[i])) <= 0) {
            stk.pop_back();
        }
        stk.push_back(points[i]);
    }
    stk.pop_back();
    return stk;
}
```

凸多边形

```
struct Convex : public Polygon {
    using Polygon::Polygon;
};

bool is_in(const Point &pt, const Convex &convex) {
    if(convex.size() == 1) {
        return pt == convex.pts[0];
    }
    if(convex.size() == 2) {
        return is_on(pt, Lineseg(convex.pts[0], convex.pts[1]));
    }
    auto check = [](const Point &a, const Point &b, const Point &c, const Point &p) {
        ld c1 = cross(b - a, p - a), c2 = cross(c - b, p - b), c3 = cross(a - c, p - c);
        return (sign(c1) != -1 && sign(c2) != -1 && sign(c3) != -1)
            || (sign(c1) != 1 && sign(c2) != 1 && sign(c3) != 1);
    };
    int n = convex.size();
    Point pivot = convex.pts[0];
    if((sign(cross(convex.pts[1] - pivot, pt - pivot)) == -1)
        || (sign(cross(convex.pts[n - 1] - pivot, pt - pivot)) == 1)) {
        return false;
    }
    int l = 1, r = n - 1;
    while(l + 1 < r) {
        int mid = (l + r) >> 1;
        if(sign(cross(convex.pts[mid] - pivot, pt - pivot)) != -1) {
            l = mid;
        } else {
            r = mid;
        }
    }
    int nxt = (l == n - 1) ? 1 : l + 1;
    return check(pivot, convex.pts[l], convex.pts[nxt], pt);
}

pair<Point, Point> tangent(const Point &pt, const Convex &convex) {
    if(is_in(pt, convex)) throw runtime_error("pt is in the convex");
    int n = convex.size();
    auto peak = [&](int l, int r, bool find_r) {
        while(l < r - 1) {
            int mid = (l + r) / 2;
            if(find_r ^ (to_left(convex.pts[(mid + n - 1) % n] - pt, convex.pts[mid] - pt) == 1)) {
                r = mid;
            } else {
                l = mid;
            }
        }
        return l;
    };
    if(to_left(convex.pts[0] - pt, convex.pts[1] - pt) == 0) {
        int idx = peak(2, n, cmp(dist(convex.pts[0], pt), dist(convex.pts[1], pt)) == -1);
        return { convex.pts[0], convex.pts[idx] };
    }
    bool all_left = true, all_right = true;
    auto fun = [&](int x) {
```

```

        if(x == 1) all_right = false;
        if(x == -1) all_left = false;
    };
    fun(to_left(convex.pts[0] - pt, convex.pts[1] - pt));
    fun(to_left(convex.pts[0] - pt, convex.pts[n - 1] - pt));
    if(all_left || all_right) {
        int idx = peak(1, n, all_left);
        return { convex.pts[0], convex.pts[idx] };
    }
    int l = 1, r = n;
    while(l < r - 1) {
        int mid = (l + r) / 2;
        if(to_left(convex.pts[0] - pt, convex.pts[mid] - pt) == 1) {
            l = mid;
        } else {
            r = mid;
        }
    }
    int split = l;
    bool flag = (to_left(convex.pts[0] - pt, convex.pts[1] - pt) == 1);
    int i1 = peak(0, split + 1, flag), i2 = peak(split, n, !flag);
    return { convex.pts[i1], convex.pts[i2] };
}

pair<Point, Point> tangent(const Line &ln, const Convex &convex) {
    int n = convex.size();
    if(n == 1) {
        return { convex.pts[0], convex.pts[0] };
    }
    if(n == 2) {
        return { convex.pts[0], convex.pts[1] };
    }

    // Direction perpendicular to the line
    Vector perp = Vector(-ln.v.y, ln.v.x);

    // Find extreme points using ternary search approach
    auto find_extreme = [&](bool find_max) -> int {
        int l = 0, r = n - 1;
        while(r - l > 2) {
            int m1 = l + (r - l) / 3;
            int m2 = r - (r - l) / 3;
            ld dot1 = dot(convex.pts[m1], perp);
            ld dot2 = dot(convex.pts[m2], perp);
            if(find_max ? (cmp(dot1, dot2) == 1) : (cmp(dot1, dot2) == -1)) {
                r = m2;
            } else {
                l = m1;
            }
        }
    }

    int best = l;
    ld best_dot = dot(convex.pts[l], perp);
    for(int i = l; i <= r; ++i) {
        ld curr_dot = dot(convex.pts[i], perp);
        if(find_max ? (cmp(curr_dot, best_dot) == 1) : (cmp(curr_dot, best_dot) == -1)) {
            best = i;
        }
    }
}

```

```
        best_dot = curr_dot;
    }
}

return best;
};

int max_idx = find_extreme(true);
int min_idx = find_extreme(false);

return { convex.pts[min_idx], convex.pts[max_idx] };
}
```

半平面交

```
vector<Line> half_inter(vector<Line> lines) {
    lines.push_back({{-inf, 0.01}, {0.01, -1.01}});
    lines.push_back({{inf, 0.01}, {0.01, 1.01}});
    lines.push_back({{0.01, inf}, {-1.01, 0.01}});
    lines.push_back({{0.01, -inf}, {1.01, 0.01}});
    sort(lines.begin(), lines.end(), [](const Line &a, const Line &b) {
        int c = cmp(a.v.arg(), b.v.arg());
        if(c != 0)
            return c == -1;
        return sign(cross(a.v, b.p - a.p)) < 0;
    });
    deque<Line> dq;
    auto bad = [](const Line &l, const Line &b, const Line &c) {
        try {
            auto p = inter(b, c);
            return to_left(l, p) < 0;
        } catch(const invalid_argument &) {
            return false;
        }
    };
    for(auto &l : lines) {
        if(!dq.empty() && cmp(l.v.arg(), dq.back().v.arg()) == 0) {
            if(to_left(l, dq.back().p + dq.back().v) < 0)
                dq.back() = l;
            continue;
        }
        while(dq.size() > 1 && bad(l, dq.back(), dq[dq.size() - 2]))
            dq.pop_back();
        while(dq.size() > 1 && bad(l, dq[0], dq[1]))
            dq.pop_front();
        dq.push_back(l);
    }
    while(dq.size() > 1 && bad(dq[0], dq.back(), dq[dq.size() - 2]))
        dq.pop_back();
    while(dq.size() > 1 && bad(dq.back(), dq[0], dq[1]))
        dq.pop_front();
    vector<Line> ret(dq.begin(), dq.end());
    int m = ret.size();
    if(m < 3)
        return {};
    vector<Point> poly;
    poly.reserve(m);
    for(int i = 0; i < m; ++i) {
        try {
            poly.emplace_back(inter(ret[i], ret[(i + 1) % m]));
        } catch(const invalid_argument &) {
            return {};
        }
    }
    if(poly.size() != m)
        return {};
    return ret;
}
```

扫描线

```
struct Rectangle {
    Point p1, p2;
};

struct Event {
    int delta, l, r;
    ld y;
};

struct SegTree {
    SegTree(const vector<ld> &_xs) : nodes(4 * _xs.size()), xs(_xs) {
        int n = xs.size();
        _build(0, 0, n);
    }
    ld query() const {
        return nodes[0].len;
    }
    // 注意左闭右开
    void update(int l, int r, int v) {
        _update(0, l, r, v);
    }
private:
    struct Node {
        int l, r, cnt;
        ld len;
    };
    vector<Node> nodes;
    vector<ld> xs;
    void _pushup(int u) {
        if(nodes[u].cnt > 0) {
            nodes[u].len = xs[nodes[u].r] - xs[nodes[u].l];
        } else if(nodes[u].r - nodes[u].l == 1) {
            nodes[u].len = 0.01;
        } else {
            int lson = (u << 1) + 1, rson = (u << 1) + 2;
            nodes[u].len = nodes[lson].len + nodes[rson].len;
        }
    }
    void _build(int u, int l, int r) {
        nodes[u] = { l, r, 0, 0.01 };
        if(r - l > 1) {
            int mid = (l + r) >> 1, lson = (u << 1) + 1, rson = (u << 1) + 2;
            _build(lson, l, mid);
            _build(rson, mid, r);
        }
    }
    void _update(int u, int l, int r, int v) {
        if(nodes[u].l >= r || nodes[u].r <= l) return;
        if(nodes[u].l >= l && nodes[u].r <= r) {
            nodes[u].cnt += v;
            _pushup(u);
            return;
        }
        int lson = (u << 1) + 1, rson = (u << 1) + 2;
        _update(lson, l, r, v);
```

```

        _update(rson, l, r, v);
        _pushup(u);
    }
};

ld scanline(const vector<Rectangle> &rects) {
    int n = rects.size();
    vector<ld> xs(2 * n);
    vector<Event> events(2 * n);
    for(int i = 0; i < n; ++i) {
        xs[2 * i] = rects[i].p1.x;
        xs[2 * i + 1] = rects[i].p2.x;
    }
    sort(xs.begin(), xs.end());
    xs.erase(unique(xs.begin(), xs.end()), xs.end());
    auto getid = [&](ld x) -> int {
        return lower_bound(xs.begin(), xs.end(), x) - xs.begin();
    };
    for(int i = 0; i < n; ++i) {
        events[2 * i] = { 1, getid(rects[i].p1.x), getid(rects[i].p2.x), rects[i].p1.y };
        events[2 * i + 1] = { -1, getid(rects[i].p1.x), getid(rects[i].p2.x), rects[i].p2.y };
    }
    sort(events.begin(), events.end(), [](const Event &a, const Event &b) {
        return cmp(a.y, b.y) == -1;
    });
    SegTree seg(xs);
    ld lasty = events[0].y;
    ld ans = 0.01;
    for(auto &e : events) {
        ans += seg.query() * (e.y - lasty);
        seg.update(e.l, e.r, e.delta);
        lasty = e.y;
    }
    return ans;
}

```

旋转卡壳

```

ld farthest_dist(const vector<Point> &pts) {
    int n = pts.size();
    auto hull = andrew(pts);
    int m = hull.size();
    int j = 1;
    ld ans = 0;
    for(int i = 0; i < m; ++i) {
        int u = (i + 1) % m;
        while(cross(hull[i], hull[u], hull[(j + 1) % m]) > cross(hull[i], hull[u], hull[j])) {
            j = (j + 1) % m;
        }
        ans = max({ ans, (hull[i] - hull[j]).len2(), (hull[u] - hull[j]).len2() });
    }
    return ans;
}

```

Pick定理

```
ll points_inside(const Polygon &poly) {
    ll twos = poly.twice_area();
    ll border = 0;
    for(int i = 0; i < poly.size(); ++i) {
        int j = (i + 1) % poly.size();
        Vector v = poly.pts[j] - poly.pts[i];
        border += gcd(abs(v.x), abs(v.y));
    }
    return (twos - border + 2) / 2;
}
```

阿波罗尼斯圆

```
// PA / PB = lambda
Circle apollonius(const Point &a, const Point &b, ld lambda) {
    if(cmp(lambda, 1) == 0) throw runtime_error("lambda is 1");
    if(lambda > 1) return apollonius(b, a, 1.01 / lambda);
    Vector v = b - a;
    Point p1 = a + v / (1 + lambda) * lambda, p2 = a - v / (1 - lambda) * lambda;
    return Circle((p1 + p2) / 2, dist(p1, p2) / 2);
}
```


圆的反演

```
// (A invert to C)'s shape, true is circle, false is line
bool inv_shape(const Circle &c, const Circle &a) {
    return cmp(dist(c.c, a.c), a.r) != 0;
}

bool inv_shape(const Circle &c, const Line &l) {
    return sign(cross(c.c - l.p, l.v)) != 0;
}

Point inverse(const Circle &c, const Point &a) {
    Vector v = a - c.c;
    return c.c + v * (c.r * c.r / v.len());
}

Circle invcircle(const Circle &c, const Circle &a) {
    if(!inv_shape(c, a)) throw runtime_error("result type mismatch");
    Vector v = a.c - c.c;
    Point p1 = a.c - v / v.len() * a.r, p2 = a.c + v / v.len() * a.r;
    Point p3 = inverse(c, p1), p4 = inverse(c, p2);
    return Circle((p3 + p4) / 2, dist(p3, p4) / 2);
}

Line invline(const Circle &c, const Circle &a) {
    if(inv_shape(c, a)) throw runtime_error("result type mismatch");
    Vector v = a.c - c.c;
    Point pt = c.c + v * 2;
    Point ip = inverse(c, pt);
    return Line(ip, Vector(-v.y, v.x));
}

Circle invcircle(const Circle &c, const Line &l) {
    if(!inv_shape(c, l)) throw runtime_error("result type mismatch");
    Point p1 = l.p, p2 = l.p + l.v, p3 = l.p - l.v;
    Point q1 = inverse(c, p1), q2 = inverse(c, p2), q3 = inverse(c, p3);
    ld x1 = q1.x, y1 = q1.y;
    ld x2 = q2.x, y2 = q2.y;
    ld x3 = q3.x, y3 = q3.y;
    ld A = x1 * (y2 - y3) + x2 * (y3 - y1) + x3 * (y1 - y2);
    ld S1 = q1.len2(), S2 = q2.len2(), S3 = q3.len2();
    ld ux = (S1 * (y2 - y3) + S2 * (y3 - y1) + S3 * (y1 - y2)) / (2 * A);
    ld uy = (S1 * (x3 - x2) + S2 * (x1 - x3) + S3 * (x2 - x1)) / (2 * A);
    Point o(ux, uy);
    ld r = dist(o, q1);
    return Circle(o, r);
}

Line invline(const Circle &c, const Line &l) {
    if(inv_shape(c, l)) throw runtime_error("result type mismatch");
    return l;
}
```

三维计算几何

```
constexpr ld eps = 1e-9l, pi = 3.14159265358979323846264338327950288l;
int sign(ld x) {
    return x > eps ? 1 : x < -eps ? -1 : 0;
}
int cmp(ld x, ld y) {
    return sign(x - y);
}

struct Point {
    ld x, y, z;
    Point() : x(0), y(0), z(0) {}
    Point(ld _x, ld _y, ld _z) : x(_x), y(_y), z(_z) {}
    Point operator+(const Point &p) const {return Point(x + p.x, y + p.y, z + p.z);}
    Point operator-(const Point &p) const {return Point(x - p.x, y - p.y, z - p.z);}
    Point operator*(ld u) const {return Point(x * u, y * u, z * u);}
    friend Point operator*(ld u, const Point &p) {return p * u;}
    Point operator/(ld u) const {return Point(x / u, y / u, z / u);}
    ld len() const {return sqrt(x * x + y * y + z * z);}
    ld len2() const {return x * x + y * y + z * z;}
    bool operator==(const Point &p) const {return cmp(x, p.x) == 0 && cmp(y, p.y) == 0 && cmp(z, p.z) == 0;}
};
using Vector = Point;
ld dot(const Vector &a, const Vector &b) {
    return a.x * b.x + a.y * b.y + a.z * b.z;
}
Vector cross(const Vector &a, const Vector &b) {
    return Vector(a.y * b.z - a.z * b.y, a.z * b.x - a.x * b.z, a.x * b.y - a.y * b.x);
}
Vector cross(const Point &o, const Point &a, const Point &b) {
    return cross(a - o, b - o);
}
ld mix_prod(const Vector &a, const Vector &b, const Vector &c) {
    return dot(cross(a, b), c);
}
ld dist(const Point &a, const Point &b) {
    return (a - b).len();
}
bool same_plane(const Point &a, const Point &b, const Point &c, const Point &d) {
    return sign(mix_prod(b - a, c - a, d - a)) == 0;
}
bool parallel(const Vector &a, const Vector &b) {
    Vector v = cross(a, b);
    return sign(v.len()) == 0;
}
ld operator/(const Vector &a, const Vector &b) {
    if(!parallel(a, b)) throw runtime_error("a and b don't parallel");
    return sign(b.x) == 0 ? a.y / b.y : a.x / b.x;
}

struct Line {
    Point p, v;
    Line() {}
    Line(const Point &p, const Vector &v) : p(p), v(v) {}
};
bool parallel(const Line &a, const Line &b) {
```

```

        parallel(a.v, b.v);
    }
    bool same_plane(const Line &a, const Line &b) {
        return same_plane(a.p, a.p + a.v, b.p, b.p + b.v);
    }
    bool is_inter(const Line &a, const Line &b) {
        return same_plane(a, b) && !parallel(a, b);
    }
    Point inter(const Line &a, const Line &b) {
        if(!is_inter(a, b)) throw runtime_error("a and b don't intersect");
        Vector w = b.p - a.p, v1v2 = cross(a.v, b.v);
        Vector ww2 = cross(w, b.v);
        ld den = v1v2.len2();
        ld t = dot(ww2, v1v2) / den;
        return a.p + a.v * t;
    }

    struct Plane {
        Point p, norm;
        Plane() {}
        Plane(const Point &p, const Vector &n) : p(_p), norm(_n) {}
        Plane(const Point &a, const Point &b, const Point &c) : p(a), norm(cross(b - a, c - a)) {}
    };
    bool parallel(const Plane &a, const Plane &b) {
        return parallel(a.norm, b.norm);
    }
}

```

字符串

KMP

```
// 得到用于KMP的数组
vector<int> kmp_f(const string &s) {
    int n = s.size();
    vector<int> f(n, 0);
    for(int i = 1, j = 0; i < n; ++i) {
        while(j > 0 && s[i] != s[j]) {
            j = f[j - 1];
        }
        if(s[i] == s[j]) {
            ++j;
        }
        f[i] = j;
    }
    return f;
}

// 寻找第一次在off后面haystack中的needle
int kmp_find(const string &haystack, const string &needle, int off = 0) {
    int n = needle.size(), m = haystack.size();
    vector<int> f = kmp_f(needle);
    for(int i = 0, j = off; j < m; j++) {
        if(haystack[j] == needle[i]) {
            ++i;
            ++j;
            if(i == n) {
                return j - n;
            }
        } else {
            if(i == 0) {
                ++j;
            } else {
                i = f[i - 1];
            }
        }
    }
    return -1;
}

// 返回haystack中有多少个needle, 可重 (即10101有两个101)
int kmp_count(const string &haystack, const string &needle) {
    int n = needle.size(), m = haystack.size();
    int ret = 0;
    vector<int> f = kmp_f(needle);
    for(int i = 0, j = 0; j < m; j++) {
        if(haystack[j] == needle[i]) {
            ++i;
            ++j;
            if(i == n) {
                ++ret;
                i = f[i - 1];
            }
        } else {
            if(i == 0) {
                ++j;
            }
        }
    }
    return ret;
}
```

```

        } else {
            i = f[i - 1];
        }
    }
}
return ret;
}

```

Z函数（扩展KMP）

```

vector<int> zfn(const string &s) {
    int n = s.size();
    vector<int> z(n);
    for(int i = 1, l = 0, r = 0; i < n; ++i) {
        if(i <= r && z[i - l] < r - i + 1) {
            z[i] = z[i - l];
        } else {
            z[i] = max(0, r - i + 1);
            while(i + z[i] < n && s[z[i]] == s[i + z[i]]) {
                ++z[i];
            }
        }
        if(i + z[i] - 1 > r) {
            l = i;
            r = i + z[i] - 1;
        }
    }
    return z;
}

```

字典树

```
struct Trie {
    Trie() : nodes(1) {}
    void insert(const string &str) {
        int rt = 0;
        for(int i = 0; i < str.size(); ++i) {
            if((nodes[rt].nxt[getnum(str[i])]) == -1) {
                nodes[rt].nxt[getnum(str[i])] = nodes.size();
                nodes.emplace_back();
            }
            nodes[rt].cnt++;
            rt = nodes[rt].nxt[getnum(str[i])];
        }
        nodes[rt].cnt++;
        nodes[rt].end = true;
    }
    bool find(const string &str) const {
        int rt = 0;
        for(int i = 0; i < str.size(); ++i) {
            if((nodes[rt].nxt[getnum(str[i])]) == -1) {
                return false;
            }
            rt = nodes[rt].nxt[getnum(str[i])];
        }
        return nodes[rt].end;
    }
    int find_prefix(const string &str) const {
        int rt = 0;
        for(int i = 0; i < str.size(); ++i) {
            if((nodes[rt].nxt[getnum(str[i])]) == -1) {
                return 0;
            }
            rt = nodes[rt].nxt[getnum(str[i])];
        }
        return nodes[rt].cnt;
    }
}

private:
    struct Node {
        array<int, 65> nxt;
        bool end;
        int cnt;
        Node() : end(false), cnt(0) {
            nxt.fill(-1);
        }
    };
    vector<Node> nodes;
    static int getnum(char x) {
        if(x >= 'A' && x <= 'Z') {
            return x - 'A';
        } else if(x >= 'a' && x <= 'z') {
            return x - 'a' + 26;
        } else {
            return x - '0' + 52;
        }
    }
}
```

```
}  
};
```

Manacher

```
string manacher(const string &_s) {  
    string s = "$";  
    for(char ch : _s) {  
        s += ch;  
        s += '$';  
    }  
    int n = s.size();  
    int max_right = 0;  
    int center = 0;  
    vector<int> dp(n, 0);  
    auto expand = [&](int left, int right) {  
        while(left >= 0 && right < n && s[left] == s[right]) {  
            --left;  
            ++right;  
        }  
        return (right - left) / 2;  
    };  
    for(int i = 0; i < n; ++i) {  
        int mirror = 2 * center - i;  
        if(i >= max_right) {  
            dp[i] = expand(i, i);  
            max_right = i + dp[i];  
            center = i;  
        } else if(dp[mirror] == max_right - i) {  
            dp[i] = expand(i - dp[mirror], i + dp[mirror]);  
            max_right = i + dp[i];  
            center = i;  
        } else {  
            dp[i] = min(dp[mirror], max_right - i);  
        }  
    }  
    auto it = max_element(dp.begin(), dp.end());  
    int id = it - dp.begin(), len = *it;  
    string ret;  
    for(int i = id - len + 1; i < id + len; ++i) {  
        if(s[i] != '$') {  
            ret += s[i];  
        }  
    }  
    return ret;  
}
```

AC自动机

千万不要忘了build!

```

struct ACAM {
    ACAM() : nodes(1) {}
    void insert(const string &str) {
        int now = 0;
        for(char ch : str) {
            int id = ch - 'a';
            int v = nodes[now].nxt1[id];
            if(v == -1) {
                v = nodes[now].nxt1[id] = nodes[now].nxt2[id] = nodes.size();
                nodes.emplace_back();
            }
            now = v;
        }
        nodes[now].cnt++;
    }
    // 不优化的，对于某些特定的题需要
    // void build() {
    //     queue<int> q;
    //     nodes[0].fail = 0;
    //     for(int i = 0; i < 26; ++i) {
    //         if(nodes[0].nxt[i] != -1) {
    //             int v = nodes[0].nxt[i];
    //             nodes[v].fail = 0;
    //             q.push(v);
    //         }
    //     }
    //     while(!q.empty()) {
    //         int u = q.front();
    //         q.pop();
    //         for(int i = 0; i < 26; ++i) {
    //             int v = nodes[u].nxt[i];
    //             if(v == -1) {
    //                 continue;
    //             }
    //             int f = nodes[u].fail;
    //             while(f != 0 && nodes[f].nxt[i] == -1) {
    //                 f = nodes[f].fail;
    //             }
    //             nodes[v].fail = max(0, nodes[f].nxt[i]);
    //             q.push(v);
    //         }
    //     }
    // }
    // int query(const string &str) const {
    //     vector<int> cnter(nodes.size(), 0);
    //     int now = 0;
    //     for(char ch : str) {
    //         int id = ch - 'a';
    //         while(now != 0 && nodes[now].nxt[id] == -1) {
    //             now = nodes[now].fail;
    //         }
    //         now = max(nodes[now].nxt[id], 0);
    //         int v = now;
    //         while(v != 0) {
    //             cnter[v] = nodes[v].cnt;
    //             v = nodes[v].fail;
    //         }
    //     }

```



```

//     }
//     int ans = 0;
//     for(int i : cnter) {
//         ans += i;
//     }
//     return ans;
// }
void build() {
    queue<int> q;
    nodes[0].fail = 0;
    for(int i = 0; i < 26; ++i) {
        int v = nodes[0].nxt2[i];
        if(v == -1) {
            nodes[0].nxt2[i] = 0;
        } else {
            nodes[v].fail = 0;
            q.push(v);
        }
    }
    while(!q.empty()) {
        int now = q.front();
        q.pop();
        for(int i = 0; i < 26; ++i) {
            int v = nodes[now].nxt2[i];
            if(v == -1) {
                nodes[now].nxt2[i] = nodes[nodes[now].fail].nxt2[i];
            } else {
                nodes[v].fail = nodes[nodes[now].fail].nxt2[i];
                q.push(v);
            }
        }
    }
    q.push(0);
    while(!q.empty()) {
        int u = q.front();
        q.pop();
        for(int i = 0; i < 26; ++i) {
            int v = nodes[u].nxt1[i];
            if(v == -1) {
                continue;
            }
            q.push(v);
            int fu = nodes[u].fail;
            if(fu != 0 && nodes[fu].cnt == 0) {
                nodes[u].fail = nodes[fu].fail;
            }
        }
    }
}

int find(const string &s) const {
    int now = 0;
    for(char ch : s) {
        int v = nodes[now].nxt1[ch - 'a'];
        if(v == -1) {
            return -1;
        }
        now = v;
    }
}

```

```

    }
    return now;
}

vector<int> query(const string &str) const {
    int now = 0;
    vector<int> ret(nodes.size(), 0);
    for(char ch : str) {
        int id = ch - 'a';
        now = nodes[now].nxt2[id];
        int u = now;
        while(u != 0) {
            ret[u]++;
            u = nodes[u].fail;
        }
    }
    return ret;
}

int size() const {
    return nodes.size();
}

private:
    struct Node {
        int cnt;
        int fail;
        array<int, 26> nxt1, nxt2;
        Node() : cnt(0), fail(0) {
            nxt1.fill(-1);
            nxt2.fill(-1);
        }
    };
    vector<Node> nodes;
};

```

后缀数组

```
vector<int> suffix_array(const string &str) {
    int n = str.size();
    int m = 128;
    vector<int> rk(n * 3, -1), sa(n), id(n), cnt(max(n, m), 0);
    int p = 0;
    for(int i = 0; i < n; ++i) {
        rk[i] = str[i];
        cnt[rk[i]]++;
    }
    for(int i = 1; i < m; ++i) {
        cnt[i] += cnt[i - 1];
    }
    for(int i = n - 1; i >= 0; --i) {
        cnt[rk[i]]--;
        sa[cnt[rk[i]]] = i;
    }
    for(int w = 1; ; w <= 1, m = p + 1) {
        int cur = 0;
        for(int i = n - w; i < n; ++i) {
            id[cur] = i;
            ++cur;
        }
        for(int i = 0; i < n; ++i) {
            if(sa[i] >= w) {
                id[cur] = sa[i] - w;
                ++cur;
            }
        }
        cnt.assign(max(n, m), 0);
        for(int i = 0; i < n; ++i) {
            cnt[rk[i]]++;
        }
        for(int i = 1; i < m; ++i) {
            cnt[i] += cnt[i - 1];
        }
        for(int i = n - 1; i >= 0; --i) {
            cnt[rk[id[i]]]--;
            sa[cnt[rk[id[i]]]] = id[i];
        }
        p = 0;
        vector<int> oldrk = rk;
        rk[sa[0]] = 0;
        for(int i = 1; i < n; ++i) {
            if(oldrk[sa[i]] != oldrk[sa[i - 1]] || oldrk[sa[i] + w] != oldrk[sa[i - 1] + w]) {
                ++p;
            }
            rk[sa[i]] = p;
        }
        if(p == n - 1) {
            break;
        }
    }
}
```

```
    return sa;
}
```

Height数组

```
vector<int> height(const string &str) {
    int n = str.size();
    vector<int> sa = suffix_array(str), rk(n), h(n);
    for(int i = 0; i < n; ++i) {
        rk[sa[i]] = i;
    }
    for(int i = 0, k = 0; i < n; ++i) {
        if(rk[i] == 0) {
            continue;
        }
        if(k > 0) {
            --k;
        }
        while(str[i + k] == str[sa[rk[i] - 1] + k]) {
            ++k;
        }
        h[rk[i]] = k;
    }
    return h;
}
```

后缀自动机

```
struct SAM {
    SAM() : nodes(1, Node(-1, 0)), last(0) {}
    void insert(char ch) {
        const int id = ch - 'a';
        const int cur = nodes.size();
        nodes.emplace_back(-1, nodes[last].len + 1);
        int p = last;
        while(p != -1 && nodes[p].nxt[id] == -1) {
            nodes[p].nxt[id] = cur;
            p = nodes[p].link;
        }
        if(p == -1) {
            nodes[cur].link = 0;
        } else {
            int q = nodes[p].nxt[id];
            if(nodes[p].len + 1 == nodes[q].len) {
                nodes[cur].link = q;
            } else {
                int clone = nodes.size();
                nodes.emplace_back(nodes[q].link, nodes[p].len + 1);
                nodes[clone].nxt = nodes[q].nxt;
                while(p != -1 && nodes[p].nxt[id] == q) {
                    nodes[p].nxt[id] = clone;
                    p = nodes[p].link;
                }
                nodes[q].link = nodes[cur].link = clone;
            }
        }
        last = cur;
        ends.push_back(cur);
    }
}

// 洛谷P3804 求出 S 的所有出现次数不为 1 的子串的出现次数乘上该子串长度的最大值。
ll solve() const {
    int n = nodes.size();
    vector<int> indeg(n, 0), topoorder;
    topoorder.reserve(n);
    for(int i = 0; i < n; ++i) {
        if(nodes[i].link != -1) {
            indeg[nodes[i].link]++;
        }
    }
    queue<int> q;
    for(int i = 0; i < n; ++i) {
        if(indeg[i] == 0) {
            q.push(i);
        }
    }
    while(!q.empty()) {
        int u = q.front();
        q.pop();
        topoorder.push_back(u);
        if(nodes[u].link != -1) {
            indeg[nodes[u].link]--;
            if(indeg[nodes[u].link] == 0) {
                q.push(nodes[u].link);
            }
        }
    }
}
```

```

        }
    }
}

ll ans = 0;
vector<ll> occur(n, 0);
for(int e : ends) {
    occur[e] = 1;
}
for(int i : topoorder) {
    if(nodes[i].link != -1) {
        occur[nodes[i].link] += occur[i];
    }
    if(occur[i] != 1) {
        ans = max(ans, occur[i] * nodes[i].len);
    }
}
return ans;
}

private:
struct Node {
    array<int, 26> nxt;
    int link;
    int len;
    Node(int lk, int ln) : link(lk), len(ln) {
        nxt.fill(-1);
    }
};
vector<Node> nodes;
vector<int> ends;
int last;
};

```

序列自动机

```
struct SeqAM {
    explicit SeqAM(const string &s) : n(s.size()), nxt(s.size() + 2, [&] {
        array<int, 26> ret;
        ret.fill(s.size() + 1);
        return ret;
    })() {
        for(int i = n - 1; i >= 0; --i) {
            nxt[i] = nxt[i + 1];
            nxt[i][s[i] - 'a'] = i + 1;
        }
    }
    bool match(const string &t) const {
        int now = 0;
        for(char c : t) {
            now = nxt[now][c - 'a'];
        }
        return now != (n + 1);
    }
private:
    int n;
    vector<array<int, 26>> nxt;
};
```

回文自动机

```
struct PAM {
    PAM() : s("#"), last(1) {
        nodes.emplace_back(-1, 0);
        nodes.emplace_back(0, 0);
    }
    int insert(char ch) {
        s += ch;
        int pos = s.size() - 1;
        int p = last, id = ch - 'a';
        while(s[pos - nodes[p].len - 1] != ch) {
            p = nodes[p].fail;
        }
        if(nodes[p].nxt[id] == -1) {
            int cur = nodes.size();
            nodes[p].nxt[id] = cur;
            nodes.emplace_back(nodes[p].len + 2, 0);
            if(nodes[cur].len == 1) {
                nodes[cur].fail = 1;
            } else {
                int f = nodes[p].fail;
                while(s[pos - nodes[f].len - 1] != ch) {
                    f = nodes[f].fail;
                }
                nodes[cur].fail = nodes[f].nxt[id];
            }
            if(nodes[cur].len == 1) {
                nodes[cur].cnt = 1;
            } else {
                nodes[cur].cnt = nodes[nodes[cur].fail].cnt + 1;
            }
        }
        last = nodes[p].nxt[id];
        return nodes[last].cnt;
    }
private:
    struct Node {
        int len, fail, cnt;
        array<int, 26> nxt;
        Node(int l, int f) : len(l), fail(f), cnt(0) {
            nxt.fill(-1);
        }
    };
    vector<Node> nodes;
    string s;
    int last;
};
```


字符串哈希

```
struct StringHash {
    explicit StringHash(const string &s) : p1(s.size() + 1, 0), p2(s.size() + 1, 0) {
        for(int i = 0; i < s.size(); ++i) {
            p1[i + 1] = (p1[i] * mul1 + s[i]) % modulo;
            p2[i + 1] = p2[i] * mul2 + s[i];
        }
    }
    ull substr(int l, int r) const {
        ull ret1 = (p1[r] - p1[l] * pmul1[r - l] % modulo + modulo) % modulo;
        ull ret2 = p2[r] - p2[l] * pmul2[r - l];
        return (ret1 << 3) ^ (ret1 >> 5) ^ ret2;
    }
private:
    vector<ull> p1, p2;
    static inline const ull c = (ull)chrono::steady_clock::now().time_since_epoch().count();
    static inline const ull mul1 = c % 131 + 131, mul2 = c % 13331 + 13331;
    static inline ull pmul1[maxn], pmul2[maxn];
    static inline int init = [&] {
        pmul1[0] = pmul2[0] = 1;
        for(int i = 1; i < maxn; ++i) {
            pmul1[i] = pmul1[i - 1] * mul1 % modulo;
            pmul2[i] = pmul2[i - 1] * mul2;
        }
        return 0;
    }();
};
```

Lyndon分解

```
vector<string> duval(const string &s) {
    int n = s.size();
    vector<string> ret;
    for(int i = 0; i < n; i++) {
        int j = i + 1, k = i;
        while(j < n && s[k] <= s[j]) {
            (s[k] < s[j]) ? (k = i) : (k++);
            ++j;
        }
        while(i <= k) {
            ret.push_back(s.substr(i, j - k));
            i += (j - k);
        }
    }
    return ret;
}
```

最小表示法

```
string minstr(const string &str) {
    int n = str.size(), i = 0, j = 1, k = 0;
    string s = str + str;
    while(i < n && j < n) {
        while(k < n && s[i + k] == s[j + k]) {
            ++k;
        }
        if(k == n)
            break;
        if(s[i + k] > s[j + k]) {
            i += (k + 1);
        } else {
            j += (k + 1);
        }
    }
    if(i == j) {
        ++j;
    }
    k = 0;
    return s.substr(min(i, j), n);
}
```

贪心

一组交叉直线经过所有点

```
struct Point {
    int x, y;
};

bool cross_all(const vector<Point> &points) {
    bool okx = true, oky = true;
    int xb = -1, yb = -1;
    for(auto [x, y] : points) {
        if(x != points[0].x) {
            if(yb == -1) {
                yb = y;
            } else if(yb != y) {
                okx = false;
            }
        }
        if(y != points[0].y) {
            if(xb == -1) {
                xb = x;
            } else if(xb != x) {
                oky = false;
            }
        }
    }
    return okx || oky;
}
```

二分

最长上升子序列

```
vector<int> lis(const vector<int> &nums) {
    int n = nums.size();
    vector<int> dp, pos(n), pre(n, -1);
    for(int i = 0; i < n; ++i) {
        auto it = lower_bound(dp.begin(), dp.end(), nums[i]);
        if(it == dp.end()) dp.push_back(nums[i]);
        else *it = nums[i];
        int j = it - dp.begin();
        pos[j] = i;
        if(j != 0) pre[i] = pos[j - 1];
    }
    vector<int> ret;
    for(int i = pos[dp.size() - 1]; i != -1; i = pre[i]) {
        ret.push_back(nums[i]);
    }
    reverse(ret.begin(), ret.end());
    return ret;
}
```

二分答案

```
template<class T> T binary(auto &&check, bool rev, T minv, T maxv, T eps, T delta) {
    T ans = (rev ? minv : maxv), hi = maxv, lo = minv;
    while((hi - lo) >= eps) {
        T mid = (lo + hi) / 2;
        if(check(mid)) {
            ans = mid;
            rev ? (lo = mid + delta) : (hi = mid - delta);
        } else {
            rev ? (hi = mid - delta) : (lo = mid + delta);
        }
    }
    return ans;
}
```

三分

```
int peak(const vector<int> &nums) {
    int n = nums.size();
    int l = 0, r = n - 1;
    int ll = (r - 1) / 3, rr = (r - 1) * 2 / 3;
    while(l < r) {
        if(nums[ll] < nums[rr]) {
            l = ll + 1;
        } else {
            r = rr;
        }
        ll = l + (r - l) / 3;
        rr = l + (r - l) * 2 / 3;
    }
    return l;
}
```

杂项

高精

```
struct HPINT {
    HPINT() : nums(1, 0), neg(false) {}
    // intentionally implicit, allowing HPINT a = 5
    HPINT(ll val) : neg(false) {
        if(val < 0) {
            neg = true;
            val = -val;
        }
        while(val) {
            int t = val % 10;
            nums.push_back(t);
            val /= 10;
        }
        if(nums.empty()) {
            nums = vector<int>(1, 0);
        }
    }
    // and HPINT b = "11451419198103141592653589793238462621718281828"
    HPINT(const string &s) {
        int off = 0, n = s.size();
        if(s[0] == '-') {
            off = 1;
            neg = true;
        } else {
            neg = false;
        }
        for(; off < n; ++off) {
            nums.push_back(s[off] - '0');
        }
        reverse(nums.begin(), nums.end());
    }
    explicit operator string() const {
        string ret;
        if(neg) {
            ret += '-';
        }
        for(int i = nums.size() - 1; i >= 0; --i) {
            ret += (nums[i] + '0');
        }
        return ret;
    }
    HPINT &operator+=(const HPINT &r) {
        if(neg) {
            if(r.neg) {
                _add(nums, r.nums);
            } else {
                if(_cmp(nums, r.nums) == 1) {
                    _sub(nums, r.nums);
                } else {
                    vector<int> rnums = r.nums;
                    _sub(rnums, nums);
                    nums.swap(rnums);
                }
            }
        }
    }
};
```

```

        neg = false;
    }
}
} else {
    if(r.neg) {
        if(_cmp(nums, r.nums) != -1) {
            _sub(nums, r.nums);
        } else {
            vector<int> rnums = r.nums;
            _sub(rnums, nums);
            nums.swap(rnums);
            neg = true;
        }
    } else {
        _add(nums, r.nums);
    }
}
return *this;
}

HPINT &operator--(const HPINT &r) {
    if(neg) {
        if(r.neg) {
            int c = _cmp(nums, r.nums);
            if(c == 1) {
                _sub(nums, r.nums);
            } else {
                vector<int> rnums = r.nums;
                _sub(rnums, nums);
                nums.swap(rnums);
                neg = false;
            }
        } else {
            _add(nums, r.nums);
        }
    } else {
        if(r.neg) {
            _add(nums, r.nums);
        } else {
            int c = _cmp(nums, r.nums);
            if(c == -1) {
                vector<int> rnums = r.nums;
                _sub(rnums, nums);
                nums.swap(rnums);
                neg = true;
            } else {
                _sub(nums, r.nums);
            }
        }
    }
    return *this;
}

HPINT &operator*=(const HPINT &r) {
    neg ^= r.neg;
    _mul(nums, r.nums);
    return *this;
}

friend ostream &operator<<(ostream &os, const HPINT &hp) {

```

```

        if (hp.neg) {
            os << '-';
        }
        for (int i = hp.nums.size() - 1; i >= 0; --i) {
            os << hp.nums[i];
        }
        return os;
    }

    friend istream &operator>>(istream &is, HPINT &hp) {
        char ch = is.rdbuf()->sgetc();
        while (ch < '0' || ch > '9') {
            if (ch == '-') {
                hp.neg = true;
            }
            ch = is.rdbuf()->sgetc();
        }
        string buffer;
        while (ch >= '0' && ch <= '9') {
            buffer += ch;
            ch = is.rdbuf()->sgetc();
        }
        hp.nums.resize(buffer.size());
        int n = buffer.size();
        for (int i = 0; i < n; ++i) {
            hp.nums[i] = buffer[n - i - 1] - '0';
        }
        return is;
    }

    bool operator==(const HPINT &r) const {
        return neg == r.neg && _cmp(nums, r.nums) == 0;
    }

    strong_ordering operator<=(const HPINT &r) const {
        if (neg != r.neg) {
            if (neg) return strong_ordering(-1);
            return strong_ordering(1);
        }
        int c = _cmp(nums, r.nums);
        if (neg) c *= -1;
        return strong_ordering(c);
    }

    HPINT operator+(const HPINT &r) const {
        HPINT ret = *this;
        ret += r;
        return ret;
    }

    HPINT operator-(const HPINT &r) const {
        HPINT ret = *this;
        ret -= r;
        return ret;
    }

    HPINT operator*(const HPINT &r) const {
        HPINT ret = *this;
        ret *= r;
        return ret;
    }

private:
    vector<int> nums;

```

```

bool neg;
static void _add(vector<int> &a, const vector<int> &b) {
    int carry = 0;
    for(int i = 0; i < b.size(); ++i) {
        if(i >= a.size()) {
            a.push_back(0);
        }
        a[i] += (b[i] + carry);
        carry = 0;
        if(a[i] >= 10) {
            carry = 1;
            a[i] -= 10;
        }
    }
    int i = b.size();
    while(carry != 0) {
        if(i >= a.size()) {
            a.push_back(0);
        }
        a[i] += carry;
        carry = 0;
        if(a[i] >= 10) {
            carry = 1;
            a[i] -= 10;
        }
        ++i;
    }
}

static void _sub(vector<int> &a, const vector<int> &b) {
    int borrow = 0;
    for(int i = 0; i < b.size(); ++i) {
        a[i] -= (borrow + b[i]);
        borrow = 0;
        if(a[i] < 0) {
            a[i] += 10;
            borrow = 1;
        }
    }
    int i = b.size();
    while(borrow && i < a.size()) {
        a[i] -= borrow;
        borrow = 0;
        if(a[i] < 0) {
            a[i] += 10;
            borrow = 1;
        }
        i++;
    }
    while(a.size() > 1 && a.back() == 0) {
        a.pop_back();
    }
}

static void _mul(vector<int> &a, const vector<int> &b) {
    using cd = complex<double>;
    constexpr double pi = 3.14159265358979323846264338327950288;
    int n = 1;
    while(n < (int)(a.size() + b.size())) {

```



```

        n <<= 1;
    }
    vector<cd> fa(n), fb(n);
    for(int i = 0; i < (int)a.size(); ++i) {
        fa[i] = a[i];
    }
    for(int i = 0; i < (int)b.size(); ++i) {
        fb[i] = b[i];
    }
    auto fft = [](auto &&fft, vector<cd> &f, bool invert) -> void {
        int n = f.size();
        if(n == 1)
            return;
        vector<cd> f0(n / 2), f1(n / 2);
        for(int i = 0; i < n / 2; ++i) {
            f0[i] = f[2 * i];
            f1[i] = f[2 * i + 1];
        }
        fft(fft, f0, invert);
        fft(fft, f1, invert);
        double theta = 2 * pi / n * (invert ? -1 : 1);
        cd wt = 1, w(cos(theta), sin(theta));
        for(int t = 0; t < n / 2; ++t) {
            cd u = f0[t], v = wt * f1[t];
            f[t] = u + v;
            f[t + n / 2] = u - v;
            wt *= w;
        }
    };
    fft(fft, fa, false);
    fft(fft, fb, false);
    for(int i = 0; i < n; ++i) {
        fa[i] *= fb[i];
    }
    fft(fft, fa, true);
    a.resize(n);
    int carry = 0;
    for(int i = 0; i < n; ++i) {
        int num = int(fa[i].real() / n + 0.5) + carry;
        carry = num / 10;
        a[i] = num % 10;
    }
    while(carry) {
        a.push_back(carry % 10);
        carry /= 10;
    }
    while(a.size() > 1 && a.back() == 0) {
        a.pop_back();
    }
}

// 1 gt, -1 lt, 0 eq
static int _cmp(const vector<int> &a, const vector<int> &b) {
    if(a.size() > b.size())
        return 1;
    if(a.size() < b.size())
        return -1;
    for(int n = a.size(), i = n - 1; i >= 0; --i) {

```

```

        if(a[i] > b[i])
            return 1;
        if(a[i] < b[i])
            return -1;
    }
    return 0;
}
};

```

快读

```

struct Qread {
    Qread() noexcept : state(true) {}
    template<integral T> Qread &operator>>(T &val) noexcept {
        if(!state) {
            val = 0;
            return *this;
        }
        T x = 0, f = 1;
        char ch = cin.rdbuf()->sgetc();
        while(ch < '0' || ch > '9') {
            if(ch == EOF) {
                state = false;
                val = 0;
                return *this;
            }
            if(ch == '-') {
                f = -1;
            }
            ch = cin.rdbuf()->sgetc();
        }
        while(ch >= '0' && ch <= '9') {
            x = x * 10 + ch - '0';
            ch = cin.rdbuf()->sgetc();
        }
        val = x * f;
        return *this;
    }
    explicit operator bool() const noexcept {
        return state;
    }
private:
    bool state;
};
Qread qread;

```

自定义哈希

```
struct MyHash {
    size_t operator()(ll x) const noexcept {
        x ^= (x >> 21);
        x ^= (x << 37);
        x ^= (x >> 4);
        x *= 0x27d4eb2f165667c5;
        x ^= (x >> 28);
        x *= 0x165667b19e3779f9;
        x ^= (x >> 31);
        return x ^ c;
    }
private:
    static inline size_t c = (size_t)chrono::steady_clock::now().time_since_epoch().count();
};
```